Optimization Algorithms in Industry 4.0

Juan Moreno Malo

April 11, 2026

Contents

1 Introduction **1**

2 Literature Review **3**
2.1 Analysis 3

3 Methodology **5**

4 Mathematical programming models for returnable transport items management in supply chains **7**
4.1 Introduction 8
4.2 Comparative analysis 9
4.3 Conclusions 11

5 An optimisation approach for supply chain procurement transport operational planning with 3D full-truck loading constraints and returnable transport items **13**
5.1 Introduction 14
5.2 Literature Review 15
5.3 Problem Description 23
5.4 Mathematical model 25
5.5 Matheuristic Procedure 33
5.6 Computational and comparative analysis 38
5.7 Practical application 40
5.8 Conclusions 41

6 Service Network Design for RTI Transport Optimization **43**
6.1 Introduction 43

6.2 Literature Review 43
6.3 Problem Statement 43
6.4 Model Formulation 47
6.5 Computational Experiments 54

7 Route Clustering Algorithms for Procurement Transport 61
7.1 Introduction 61
7.2 Literature Review 63
7.3 Problem Description 64
7.4 Mathematical Formulation 66
7.5 Branch and Price Solution Approach 69
7.6 Computational Experiments 73
7.7 SCRC As A Decision Support Tool 73
7.8 Conclusions and Future Research 75

8 Global Results 77

9 Conclusions 79

A Mathematical Model 81

B Full Non Linear Model 85

List of Figures

1.1 The core concept of the thesis 1

2.1 Statistics of reviewed papers (Article 1) 3

3.1 General Methodology Flowchart 5

5.1 Research Methodology 22
5.2 Product to type of reusable container 23
5.3 Enter Caption 24
5.4 Enter Caption 24
5.5 Description of the bottom corners position 30
5.6 Procceses of plant and supplier 37

6.1 Full RTIs routes 44
6.2 Product quantities per RTI 44
6.3 Full and empty folded RTIs 45
6.4 One-to-one RTI return strategy 45
6.5 SND with minimized RTI return transport cost 45

7.1 Routes of the supply chain 64
7.2 Clustered routes example 65
7.3 Clustered supply chain 66
7.4 Branch and Price 70
7.5 Transport cost vs. Efficiency parameter 74

8.1 Cross-chapter comparison 77

Chapter 1

Introduction

Welcome to the thesis. Lorem ipsum dolor sit amet, consectetuer adipiscing elit.

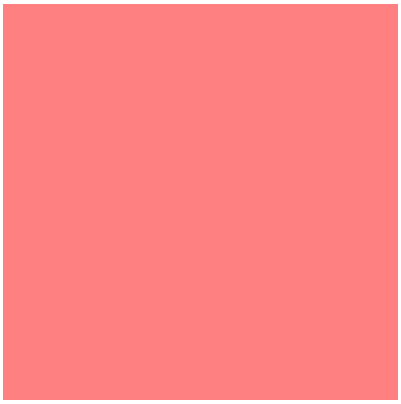


Figure 1.1: The core concept of the thesis

Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla.

Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Chapter 2

Literature Review

2.1 Analysis

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris. As seen

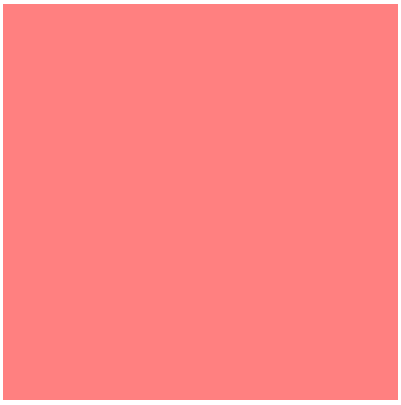


Figure 2.1: Statistics of reviewed papers (Article 1)

in Smith, 2023...

Chapter 3

Methodology

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

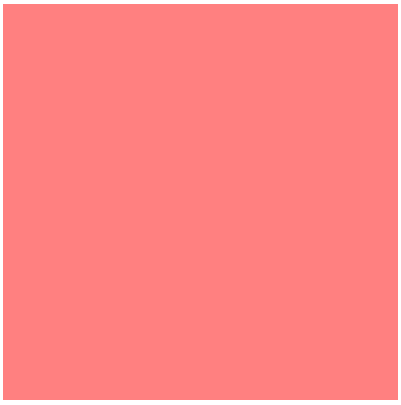


Figure 3.1: General Methodology Flowchart

Chapter 4

Mathematical programming models for returnable transport items management in supply chains

Returnable transport items (RTIs) such as pallets, crates and containers are reusable packaging assets that circulate in closed-loop supply chains to reduce waste and improve resource utilisation. However, the management of RTIs is often disconnected from the production planning of the goods they transport. This paper analyses five representative mathematical programming models that jointly address transport and inventory decisions for finished goods and RTIs in supply chains. The models are compared in terms of modelling approach, objectives, decision level, network structure, transportation and solution methods. The analysis reveals that while existing models integrate inventory and routing decisions for RTIs at the tactical and operational levels, the explicit incorporation of RTI requirements into supply chain material requirements planning (MRP) logic remains an open research gap. The identified research gap serve as the basis for a future model that integrates RTI planning into MRP within a collaborative supply chain structure.

Keywords— Returnable transport items mathematical programming supply chain collaborative logistics

4.1 Introduction

In modern supply chains, raw materials, components and finished products flow through multiple echelons accompanied by their packaging. Transport packaging — commonly referred to as returnable transport items (RTIs) — includes pallets, crates, containers and boxes that are designed to be reused Mahmoudi and Parviziomran, 2020. Unlike disposable packaging, RTIs circulate in closed-loop supply chains (CLSCs) Govindan et al., 2015, where loaded RTIs are delivered to customers in forward flows and empty RTIs are collected and returned in reverse flows for cleaning, inspection and reuse Hellström and Johansson, 2010. Using RTIs can reduce packaging waste, lower CO_2 emissions over the packaging lifecycle, and decrease total logistics costs once effectively managed Glock, 2017.

Despite these benefits, the management of RTIs presents complexity in terms of coordination of forward and reverse flows, inventory management at multiple echelons, procurement of new units to replace losses, and transportation planning for both loaded or full and empty items Mahmoudi and Parviziomran, 2020. A breakdown in the supply of RTIs directly impacts the flow of finished products or raw materials — leading to increased delivery times, backlogging and storage costs Iassinovskaia et al., 2017. Yet in practice, RTI planning is typically managed independently from production planning systems. In traditional material requirements planning (MRP), demand for components and raw materials is derived from the master production schedule (MPS) through the bill of materials (BOM) Mula et al., 2010. RTIs, despite being a necessary input to pack and ship products, are usually excluded from this logic.

A promising approach to mitigate these issues is the adoption of collaborative strategies, where RTIs are shared across supply chain partners through pooling arrangements rather than managed in dedicated closed loops by individual firms. Comparative studies in the automotive industry have shown that shared RTI management modes can reduce total supply chain costs compared to dedicated modes Q. Zhang et al., 2015.

The purpose of this paper is to identify and analyse five mathematical programming models that address RTI management in supply chains. The models are compared according to their modelling approach, objectives, decision level (strategic, tactical or operational), network structure, transportation decisions, vehicle capacity modelling, collaborative aspects and solution methods.

The remainder of this paper is structured as follows. Section 4.2 presents the comparative analysis. Section 4.3 offers the conclusions and future research directions.

4.2 Comparative analysis

Five mathematical programming models were selected as references to develop a future optimisation model that integrates RTI management into production planning within a collaborative supply chain.

Sarkar et al. Sarkar et al., 2017 propose a mixed-integer non-linear programming (MINLP) model for a multi-echelon CLSC comprising a single manufacturer, multiple retailers and a third-party logistics (3PL) provider. The 3PL owns the RTIs and handles all freight in both directions: outbound delivery of finished products packed in RTIs to retailers, and inbound collection of end-of-life products to remanufacture, along with the return of emptied RTIs. The network follows a one-to-many pattern with direct sequential shipments. Transportation cost is computed per container and per unit distance. The model operates at the tactical decision level, determining cycle time, remanufacturing rate, container capacity and the retailer shipment sequence. Carbon emission costs from both forward and reverse flows are incorporated. An improved algorithm combines analytical optimisation for continuous variables with enumeration for discrete ones. Numerical experiments demonstrate that ignoring transportation and carbon emission costs changes the remanufacturing rate from the optimal 42% to 72%, increasing total costs.

Liu et al. Liu et al., 2020 develop a mixed-integer programming (MIP) model to optimise the distribution and routing of reusable transport items from a third-party RTI pooling company to its supply chain customers. The model is based on a Chinese RTI pooler operating multiple container management centres (CMCs). The network follows a many-to-many pattern with a hub-and-spoke topology: CMCs act as regional hubs that store, inspect and maintain RTIs, while customers are served by trucks dispatched from these hubs. Vehicle capacity is expressed in volume. The model operates at both the tactical decision level (allocating customer demand across CMCs) and the operational decision level (vehicle routing). A two-stage approach first allocates demand across CMCs, then solves vehicle routing for each CMC. The key contribution is a sharing strategy that removes the dedicated assignment of customers to specific CMCs. Instead, empty RTIs returned by downstream players are stored at the nearest CMC and dispatched to any customer. Empirical results demonstrate a 28.1% reduction in transportation costs across three regions of China. However, the inbound return of empties from downstream players to CMCs is outside its scope, and therefore the model does not contemplate the manufacturing decisions of its customers or how the RTIs will be reinserted into the system.

Zhang et al. Y. Zhang et al., 2022 formulate a bi-objective mixed-integer linear programming (MILP) model for a food CLSC planning problem with RTIs. The network comprises a single manufacturer and multiple retailers, following a one-to-many pattern with direct transportation — each retailer receives its own dedicated

vehicle trips. The manufacturer ships products using a homogeneous fleet of refrigerated vans, each with a fixed capacity expressed in number of RTIs. A distinctive feature of this model is the assumption that the volume of an empty RTI can be reduced by half (e.g. through folding or nesting). Consequently, the return trip can carry up to twice as many empty RTIs as the forward trip carries loaded ones. Loaded or full RTIs travel outbound to retailers and empty RTIs are collected on the return trip of the same vehicle, constituting simultaneous pickup and delivery. The model operates at the tactical decision level with a multi-period planning horizon. The two conflicting objectives are to maximise total profit and to minimise the number of vehicles used as a proxy for carbon emissions. A kernel-search heuristic based ε-constraint method obtains approximate Pareto fronts.

Achamrah et al. Achamrah et al., 2022 propose a mathematical programming model for virtual pooling of RTIs in a two-level CLSC comprising multiple suppliers and multiple customers, following a many-to-many pattern with direct deliveries. RTIs are of various types (e.g. polyboxes, stillages). Some RTI types are collapsible, meaning their empty-state volume is smaller than their loaded-state volume, while non-collapsible types occupy the same space regardless of state. The central innovation is that suppliers with common customers coordinate to virtually pool their empty RTI inventories: when loaded RTIs are delivered to a customer, any visiting supplier may pick up empty RTIs regardless of ownership, enabling simultaneous delivery and collection. The model operates at the operational decision level, determining time period-by-period inventory quantities, transportation flows and pickup/delivery schedules. The objective minimises total costs including inventory holding, screening, maintenance, transportation, sharing and purchasing costs for new RTIs. An artificial-immune-system-based algorithm enhanced with deep reinforcement learning handles the combinatorial complexity. Computational experiments demonstrate that virtual pooling achieves cost reductions of approximately 40%.

Soysal et al. Soysal et al., 2023a address a multi-period closed-loop inventory routing problem (CIRP) in a vendor managed inventory (VMI) system. The network consists of a single vendor depot and multiple customers, following a one-to-many pattern with multi-stop vehicle routing. Vehicles depart from the depot, visit a subset of customers on a single tour delivering fully-loaded RTIs and collecting empty ones simultaneously, and return to the depot. This model features the most detailed vehicle capacity formulation among the five reviewed: loaded and empty RTIs share the same vehicle simultaneously, each occupying a different amount of space depending if they are full or empty RTIs. The total volume of loaded and empty RTIs on board must not exceed the vehicle capacity, meaning the available space is dynamically allocated between outbound deliveries and inbound collections. An empty RTI consumes approximately two-thirds of the space of a loaded one. The model operates at the tactical-operational decision level with a multi-period planning horizon. In each time period, the

vendor decides how many empty RTIs to buy, how many to collect from each customer and how many loaded RTIs to deliver. Demand uncertainty is handled through service-level constraints with normally distributed demand. A Relax-and-Fix heuristic is proposed for large instances.

Table 4.1 summarises the different analysed models.

Table 4.1: Model summary table

Ref.	SC structure	Decision level	RTI decisions	Transport mode	Fleet	Vehicle capacity model	Foldable RTI	Modelling approach	Solution procedure
Sarkar et al., 2017	Manuf. – 3PL – Retailers	T	Inv., purchase	DS	-	-	No	MINLP	Analytical + enum.
Liu et al., 2020	CMC hubs – Customers	T | O	Inv.	VRP	HM	V	N/A	MIP	CPLEX (alloc. + CVRP)
Y. Zhang et al., 2022	Manuf. – Retailers	T	Inv., purchase	DS	HM	RTI units	Yes	Bi-obj. MILP	KS ε-constr.
Achamrah et al., 2022	Suppliers – Customers	O	Inv., purchase	DS	HT	V	Yes	MIP	AIS + deep RL
Soysal et al., 2023a	Vendor – Customers	T–O	Inv., purchase, routing	VRP	HM	V	Yes	MILP	RAF

[a] SC: supply chain; 3PL: third-party logistics; CMC: container management centre; T: tactical; O: operational; DS: direct shipment; VRP: vehicle routing problem; HM: homogeneous fleet; HT: heterogeneous fleet; V: volume-based capacity. KS: kernel-search; AIS: artificial immune system; RL: reinforcement learning; RAF: Relax-and-Fix heuristic

4.3 Conclusions

The five analysed models apply mathematical programming to optimise RTI management decisions in supply chains. All five models consider inventory and transport decisions for RTIs. However, the integration of RTI management with production planning varies considerably across models.

The main identified research gap is the absence of explicit integration between RTI planning and MRP logic. In manufacturing practice, RTIs are functionally equivalent to a component in the BOM — without available RTIs, products cannot be packed and shipped, disrupting production schedules. Yet none of the reviewed models incorporates RTIs into a BOM structure or derives RTI requirements from a MPS. This gap remains an open direction for future research.

The identified gaps suggest two complementary research directions. At the operational decision level, a promising avenue is the integration of RTI closed-loop management with procurement transport planning under realistic vehicle capacity constraints, including three-dimensional loading restrictions and the volume difference between loaded or full and empty RTI states, and RTI requirements for all the supply chain nodes. At the tactical decision level, a further direction concerns the strategic design of RTI return networks across multi-node supply chains, optimising return routes, transport modes, and RTI allocation to capture the cost and environmental benefits of collaborative sharing strategies, whereby empty RTIs are consolidated and rerouted across supply chain partners rather than returned individually to their origin nodes. Together, these two main future research directions address complementary dimensions of the RTI planning problem — operational efficiency on the one hand, and tactical network design and collaborative pooling on the other.

Acknowledgements

This work was funded by the Horizon Europe Framework Programme with grant number 101177842: “Unified Modelling and Automated Scheduling for Manufacturing as a Service" (UniMaaS).

Chapter 5

An optimisation approach for supply chain procurement transport operational planning with 3D full-truck loading constraints and returnable transport items

This paper proposes a new mixed integer linear mathematical programming (MILP) model, dubbed 3D-SCPTOP, for procurement transport planning in manufacturing supply chains. The model effectively integrates procurement and transport decisions while accounting for three-dimensional full-truck loading constraints and managing returnable transport items (RTIs). Thus, we optimise the material to procure for an assembly manufacturing site, RTIs closed-loop management, and the involved transport decisions, including the truck loading configuration. Procurement transport decisions are based on the demand of assemblers, the supplier's inventory levels, and the dimensions of the truck / RTIs. The optimisation model aims to minimise transportation and inventory costs. As the 3D loading problem is NP-hard, a matheuristic (Math-3D-SCPTOP) approach is developed to obtain good quality solutions for large instances in an efficient computational time. The 3D-SCPTOP model was validated with a case study in an automo-

tive seating first-tier supplier. Regarding inventory and transport efficiency, the results of the 3D-SCPTOP model are superior to the company procedures.

Keywords— Procurement, inventory, transport planning, full-truckload, 3D container loading

5.1 Introduction

Optimizing the use of resources in manufacturing supply chains is crucial for boosting economic growth, improving social well-being, and minimizing environmental impact. Integrating transport, inventory, and production processes is a fundamental characteristic of supply chains (Stadtler et al., 2015) because joint decisions will improve supply chain performance(Díaz-Madroñero et al., 2017).

Procurement refers to the process of acquiring goods and services in a supply chain (Chopra & Meindl, 2015). One of the main goals of procurement is to obtain raw materials, parts or finished goods for a factory or distribution centre from a supplier. As mentioned in (Fleischmann et al., 2015), the responsibility for transport planning usually lies with the supplier. However, sometimes the manufacturer makes transport decisions of its supplies in, for example, the automotive industry. Furthermore, an essential asset in the automotive supply chain is not considered in traditional material requirement planning (MRP): returnable containers or returnable transport items (RTI), which are used as tertiary packaging. RTIs enable the storage and transportation of RM and FG through the supply chain. Usually, the manufacturing plant supplies empty RTIs to its supplier, while the supplier delivers full RTIs with the FG or RM. Therefore, during the procurement process, transport and inventory decisions should be integrated, including RTI management. Here, we focus on optimising the operational decisions made during procurement transport planning.

According to (Díaz-Madroñero, Peidro, & Mula, 2014), using algorithm-based methods for procurement transport planning is more effective than relying on manual approaches based on planners' intuition. Manual planning can be short-sighted and lead to less efficient outcomes. This becomes even more significant when optimizing procurement transport tasks that aim to reduce the number of trucks and inventory levels, especially under 3D loading constraints where the precise placement of containers within trucks must also be determined (cite). Furthermore, the capacity constraints used in the scientific literature on procurement transport planning are based mainly on freight volume or the total number of loaded product units (Hashemi-Amiri et al., 2023) and (Díaz-Madroñero, Peidro, & Mula, 2014). The current research seeks to address this gap by introducing a new mixed-integer linear programming (MILP) model. This model uniquely combines procurement transportation with considerations for 3D loading, inventory management, and RTI management. Hereafter, we refer to this model formulation as 3D supply chain procurement transport operational planning (3D-SCPTOP). Additionally, a matheuristic procedure (MATH-3D-SCPTOP) is researched.

In this paper, we present three main contributions: (i) formulating a novel MILP model, 3D-SCPTOP, that integrates procurement transport, RTI management and inventory decisions with 3D full-truck loading constraints; (ii) developing a matheuristic, Math-3D-SCPTOP, to solve the 3D-SCPTOP model in a reasonable time, which is to be used for real-world instances; (iii) validating and testing the 3D-SCPTOP model with a case study at a first-tier supplier of the automotive sector.

Readers are referred to the typology by (Wäscher et al., 2007) for a global view of container loading problems, where the different dimensions of the loading problem are described.

Furthermore, (Silva et al., 2019) summarise the exact methods used in the literature to solve container loading problems, and (X. Zhao et al., 2016) explore other solution approaches in the literature. Readers are also referred to the literature reviews of (Mosca et al., 2019) and (Díaz-Madroñero et al., 2015), which focus more on integrating inventory decisions and transport. Here, we jointly address container loading, procurement transport and inventory decisions. Furthermore, we refer readers () for insights about the packaging methods used in supply chains.

The paper is organised as follows. Section 2 provides a literature review related to the problem under study. Section 3 presents the problem description. Section 4 describes the 3D-SCPTOP MILP formulation and the developed matheuristic solution approach, MATH-3D-SCPTOP. Section 5 validates and experiments with the proposal in a real case study. Finally, Section 6 concludes and puts forward further research lines.

5.2 Literature Review

The goal of this research is to integrate procurement transport operational planning decisions with 3D loading restrictions and RTI packaging. Based on the authors' research, no previous work has been identified that meets this criterion. Therefore, three independent, but sometimes overlapping, knowledge bodies have been reviewed: procurement transport and inventory planning, container loading problems and RTI literature.

5.2.1 Procurement transport and inventory planning

Models that integrate transport and inventory costs during the procurement process are widely studied in the literature. As noted in (Langley et al., 2020), transportation facilitates the movement of raw materials and finished products between suppliers, warehouses, manufacturers, and retailers. Therefore, transport processes play a crucial role in supply chains. (Fleischmann et al., 2015) distinguish between two main types of transportation within a supply chain: "procurement transport," which involves moving materials from suppliers to manufacturing sites, and the "distribution of goods," which concerns delivering finished products from factories to customers. This article focuses specifically on addressing the challenges related to procurement transportation.

There are two main dimensions of procurement transport problems according to the supply chain configuration and the applied transport method. Regarding supply chain configurations (Mosca et al., 2019): (i) single source to a single destination (SS); (ii) multiple sources to a single destination (MS); (iii) single source to multiple destinations (SM); (iv) multiple sources to multiple destinations (MM). Source components determine the number of supplier locations that are taken into account, while destination components denote the number of supplied manufacturers. Additionally, different transport methods are identified in (Díaz-Madroñero, Peidro, & Mula, 2014):milk run, less-than-truckload (LTL), full-truckload (FTL) . We also analyze the formulation of the transportation capacity constraints in the scientific literature.

(Berman & Wang, 2006) address the problem of optimising distribution strategies to deliver products from suppliers to plants, whose aim is to minimise transport, pipeline and plant inventory costs by using cross docks. A mixed integer non-linear programming (MINLP) formulation, a Lagrangian relaxation heuristic, a greedy heuristic, and a branch-and-bound algorithm are proposed to model and solve the problem.

(Sancak & Salman, 2011) present a MILP model to optimise procurement transport decisions in a SS supply chain configuration using LTL and FTL transport methods. The

objective is to minimise transport and inventory holding costs, while fulfilling production plan requirements on a multiperiod planning horizon. The study explores the possibility of delaying LTL shipments to the subsequent time period, and utilising safety stocks whenever necessary, in a stochastic environment by employing data from a bus manufacturer. The findings demonstrate that delaying shipments leads to reduced holding and transport costs with a minimal stockout risk.

(Schöneberg et al., 2010) examine the concept of delivery profiles, which define the delivery frequency at a fixed rate, offering structured planning, minimizing management effort, and improving coordination in logistics. In this context, FTL transport is used when the goods' volume exceeds a vehicle's capacity, making direct shipment possible. LTL is applied when the volume is insufficient to fill a truck, and the goods are first sent to a consolidation center before reaching the final destination. In (Schöneberg et al., 2013), the proposed model accounts for demand uncertainty.

(Díaz-Madroñero, Mula, & Peidro, 2014) present a fuzzy multi-objective integer linear programming (FMOILP) model that focuses on operational procurement transport planning in an automotive supply chain, which involves a single source and a single destination while utilising FTL. The developed model aims to minimise the number of trucks employed for procurement transport and the inventory level by considering the uncertainties related to transport capacity levels.

(Hein & Almeder, 2016) investigate the benefits of integrated procurement transport and production planning for supply and production processes by comparing a sequential approach, where production is firstly determined, followed by procurement transport decisions, plus an integrated approach in which decisions are jointly made. Numerical experiments reveal that the integrated planning approach leads to substantial cost savings.

(Díaz-Madroñero et al., 2017) present an novel MILP model that integrates procurement, production and transport planning decisions within manufacturing systems. This model extends beyond traditional material requirement planning (MRP) systems by incorporating procurement transport decisions from multiple sources and a single destination into a unified optimization framework. It considers the capacities of both material and production resources, along with procurement transport planning decisions, by employing various shipping modes such as FTL, LTL, and milk-run.

(Chitsaz et al., 2019) address integrating production, inventory and procurement transport into a manufacturing context by defining this problem as an assembly routing problem (ARP). It introduces a novel MILP model for the ARP and proposes a three-phase decomposition matheuristic for its solution.

(Baller et al., 2022) examine the optimization of procurement transport in an automotive factory, focusing on first-tier suppliers at an aggregate level across multiple sources (570) to a single destination, using FTL, LTL, and courier express service (CES) as transportation methods. The proposed MILP approach includes a cyclic inventory constraint, truck utilization, and storage limitations.

(Hashemi-Amiri et al., 2023) propose an integrated approach to optimize a perishable food supply chain. The model combines supplier selection, procurement and distribution transport, production scheduling, and vehicle routing to improve supply chain efficiency and resilience. It handles demand uncertainties through a distributionally robust chance-constrained model. The model is reformulated as MILP by utilizing duality and linearization techniques.

5.2.2 Procurement Transport with RTIs

Some models further integrate the procurement and delivery of RTIs into the procurement transport literature. In this sense, RTIs are procured to the supplier, who loads the items and sends them to customer. We extend the SC definitions previously defined to contemplate the RTIs management strategies.

model a food closed-loop supply chain network design problem incorporating returnable transport items (RTIs), product perishability, facility location and demand uncertainty. A two-stage non-linear chance-constrained programming model minimizes fixed facility, production, transport, inventory holding and RTI costs while ensuring a target service level. It's linearized via sample average approximation and distribution-free methods to capture forward RTI-based shipments and reverse flows for reuse. To solve large instances, an improved Lagrangian relaxation algorithm decomposes facility-location and flow decisions, outperforming CPLEX on solution quality and speed.

(Soysal et al., 2023b) develop a tactical-level, multi-period closed-loop Vendor-Managed Inventory (VMI) system for RTIs in an agro-food SC. The SM SC integrates forward flows of full RTIs from a single vendor to multiple customers and reverse flows of empties back to the vendor for cleaning . All routing and inventory decisions are made centrally by the vendor under full information sharing. Sustainable plans in terms of amounts of energy spent and empty crates collected are devised and evaluated.

(Guo et al., 2024) model two-tier Container Management Centre (CMC) network . Empty RTIs are dispatched from central CMC hubs to upstream suppliers, loaded with products, carried by truck to downstream customers, then returned empty to regional CMCs for cleaning and storage before being shipped back to central hubs. A neural network is used for demand forecasting. A MILP is proposed that maximises total profit by optimally determining transport costs and the penalties applied to late RTIs returns.

(Y. Zhang et al., 2024) formulate a perishable-food inventory-routing MILP with multiple RTI types in a SM supply chain, integrating procurement of products and pickup of both filled and empty RTIs. Decisions are made on a operational level, optimally determining per period the quantities to pack, deliver, pick up, purchase and lose, as well as the vehicle routes in order to maximise the total profit of the SC that is equal to the total quality-dependent revenue minus the total costs involved, including inventory holding costs, product cost, transportation costs, and RTI purchasing and loss costs. The model is solved exactly via CPLEX solver and approximately via a tailored Kernel Search matheuristic.

5.2.3 Container loading

Container loading problems encompass a subset of geometric assignment problems known as cutting and packing ($C\&P$) problems. They revolve around the allocation of n-dimensional orthotope items in larger n-dimensional orthotope containers. Here the aim is to optimise an objective function while taking into account two key geometric constraints: complete containment of items inside containers and lack of item overlap. A loading pattern serves as a formal representation of a solution to such an assignment problem. Dyckhoff introduced in (Dyckhoff, 1990) a typology that unifies $C\&P$ problems using four different criteria: dimensionality of space, the objective of assignment, containers types and box type. Consequently, (Wäscher et al., 2007) develop a new typology that has become the standard in the $C\&P$ literature. We briefly describe the typology used in (Wäscher et al., 2007) based on five different criteria: (i) dimensionality of space; (ii) kind of assignment (input minimisation of containers for all small items and loading patterns; output maximisation to pack a subset of

small items with limited container availability;) (iii) assortment of small item by distinguishing if they are either a weakly (or strongly) heterogeneous assortment or identical small items ; (iv) assortment of large objects (if it is either several large objects or one large object); (v) shape of small items (circular or of another shape). Table 1 links the standard $C\&P$ nomenclature and the criteria of the typology according to (Wäscher et al., 2007).

Table 5.1: $C\&P$ nomenclature and criteria

<table>
<tr><td colspan="2">Assortment of small items</td><td>Identical</td><td colspan="2">Weakly hetereogenous</td><td colspan="2">Strongly heterogenous</td></tr>
<tr><td colspan="2">Objective</td><td>Output maximization</td><td>Input minimization</td><td>Output maximization</td><td>Input minimization</td><td>Output maximization</td></tr>
<tr><td colspan="2">Assortment of large items</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td rowspan="4">Fixed Dimensions</td><td>Identical</td><td rowspan="3">X</td><td>Single Stock Size Cutting Stock Problem (SSSCSP)</td><td>Multiple Identical Large Object Placement Problem (MILOPP)</td><td>Single Bin Size Bin Packing Problem (SBSBPP)</td><td>Multiple Identical Knapsack Problem (MIKP)</td></tr>
<tr><td>Strongly heterogeneous</td><td>Residual Cutting StockProblem (RCSP)</td><td rowspan="2">Multiple Heterogenous Large Object Placement Problem (MHLOPP)</td><td>Residual Bin Packing Problem (RBPP)</td><td rowspan="2">Multiple Heterogeneous Knapsack Problem (MHKP)</td></tr>
<tr><td>Weakly hetereogenous</td><td>Multiple Stock Size Cutting Stock Problem (MSSCSP)</td><td>Multiple Bin Size Bin Packing Problem (MBSBPP)</td></tr>
<tr><td>One large object</td><td>Identical Item Packing Problem (IIPP)</td><td>X</td><td>Single Large Object Placement Problem (SLOPP)</td><td>X</td><td>Single Knapsack Problem (SKP)</td></tr>
<tr><td>Variable dimension</td><td>One large object</td><td>X</td><td>Open Dimension Problem (ODP)</td><td>X</td><td>Open Dimension Problem (ODP)</td><td>X</td></tr>
</table>

The first mathematical programming model in a scientific journal to describe the container-loading problem was developed in (C. S. Chen et al., 1995), where the authors introduced an MILP model to address the 3D input minimisation problem by considering rotations and containers of varying dimensions. However, it does not specify the use of weakly or strongly heterogeneous boxes or containers. Additionally, real-world constraints are modelled like, for example, weight distribution.

(Junqueira et al., 2012b) propose an alternative MILP formulation to (C. S. Chen et al., 1995) using binary encoding for box position, which utilises normal patterns to reduce possible box coordinates by considering multiples of boxes dimensions and ensuring optimal solutions. Additionally, loading and cargo constraints are developed to ensure the static and dynamic stability of the cargo. Boxes must be supported by other box top faces or the container floor. The same authors integrate further loading restrictions in (Junqueira et al., 2012a), where they integrate multidrop constraints, which address the challenge of packing a truck with boxes for different customers at various destinations. The goal is to plan the container's loading to minimise the time spent unloading and reloading boxes at each destination.

(Zhu & Lim, 2012) devise a tree search-based method to solve SLOPP and SKP with 3D constraints. Their approach involves evaluating free spaces by employing a look-ahead tree search and selecting optimal placements based on the best free space-block pairs.

(Gonçalves & Resende, 2013) develop a biased random-key genetic algorithm (BRKGA) designed to solve SBSBPP. The algorithm addresses geometric constraints and introduces a maximal-space representation to handle available free spaces in bins. A novel placement

procedure is hybridised with the BRKGA, which optimises the order of box packing and the parameters utilised by the placement procedure to offer an effective solution approach to an NP hard problem.

(Bortfeldt & Wäscher, 2013) identify constraints for $C\&P$, including item-related constraints (orientation constraints, load bearing, loading priorities), container-related constraints (weight distribution and weight limit), and cargo-related constraints (complete-shipment constraints) and load-related constraints (complexity constraints or stability constraints). The study also highlights the achievements and deficits in container loading research that pertain to these constraints.

(Paquay et al., 2016) propose an MILP model for the MBSBPP. One noteworthy constraint is the assumption that containers are truncated parallelepiped instead of rectangular boxes. They also consider constraints, such as box rotation, container load bearing, weight distribution, stability and fragility of items.

(Mahvash et al., 2018) introduce a novel resolution procedure for the SBSBPP through the column generation technique, where the columns that are generated are the loading configurations of a single bin. For the pricing develop problem they develop a heuristic to find new columns with a negative reduced cost. This approach not only produces better results than the previous articles on the literature, but also gives new lower bounds.

(Silva et al., 2019) study SSLOP models with 3D constraints. These authors compare the different approaches that exist in the literature for model packing problems based on two fundamental approaches: discretised formulation (Junqueira et al., 2013) and continuous formulation (C. S. Chen et al., 1995) with subsequent modifications. They also indicate that discretised formulation(Junqueira et al., 2012a) performs the best for instances with similar sized boxes. However, it encounters memory limitations for large instances. For other sets of instances, the adapted formulations of (C. S. Chen et al., 1995) and (Paquay et al., 2016) outperform other adaptations. Further research topics suggest including the design of new formulations by considering the limited approaches in the literature.

(Kurpel et al., 2020) introduce techniques and exact approaches to solve input minimisation and output maximisation problems for container-loading problems. Their methods consider practical constraints, such as orientation, separation of boxes and vertical stability. A set partitioning method is also employed in order to obtain upper bounds in a container-loading input minimisation problems. The authors adapt and evaluate four discretisation methods found in the existing literature and incorporate mathematical formulations to address the aforementioned constraints.

(Alonso et al., 2020) develop an algorithm by the greedy randomised adaptive search procedure (GRASP) to tackle a practical container-loading problem with time-sensitive constraints. Their approach addresses dynamic stability constraints while stacking products on pallets and loading them onto trucks during different time periods. Their approach provides high-quality solutions and surpasses an integer programming formulation in efficiency and effectiveness terms.

(Ali et al., 2022) conduct a comprehensive literature review by identifying constraints and solution methods for offline problems with full knowledge of the items to be packed and those tackling the online problems, where items are packed immediately when they arrive, and no prior knowledge of an item is known. The study also identifies and classifies solution approaches for offline packing into six categories: exact methods, constructive heuristics; local search and metaheuristic methods; approximation algorithms; machine learning and hyper-heuristic methods; tree search methods. (Harrath, 2022) proposed a thee stage heuristic based on layer generation to find good quality solutions the SBSPP problem for large

instance up to 993 bins and 200000 items taking into account load balancing constraints.

(Tresca et al., 2022) develop a matheuristic to solve the SBSBPP that includes items with different IDs and different categories, to be delivered to different clients. The algorithm creates monoitem, monocategory, and mixed layers which then are used to compute the final loading configurations of the bins.

(Li et al., 2022) develope a hybrid adaptaive large neighborhood search algoritm to solve the multiple container loading problem, which is tries pack a given set of items with the minimum number of containers while maximazing the space utilization of them. Therfore, it is both a MBSBPP and a MSSCSP. Their approach takes into account some practical applications such as supsension constraints, rotation and weight limit.

(M. Chen et al., 2023) present a new approach to the 3D-BPP, introducing the 3D-BSDPP which focuses on optimizing both the design of bin sizes and the packing process. A novel Hybrid Biogeography-Based Optimization (HBBO) algorithm is proposed, integrating differential evolution and a local search strategy for enhanced performance.

One noteworthy trend in $C\&P$ is integrating loading problems with routing. This is known as capacitated vehicle-routing problem with 3D or 2D loading constraints (3L-CVRP or 2L-CVRP). (Junqueira et al., 2012b) present a MILP model with binary encodings for 3L-CVRP by including loading, cargo and multidrop constraints. The combination of these two NP-hard problems requires using heuristics and metaheuristics for real-world instances, as seen in (Dominguez et al., 2016), (Alonso et al., 2022) and (Wei et al., 2015). Another noteworthy trend is to use reinforcement learning techniques to solve $C\&P$, as in (Que et al., 2023) and (H. Zhao et al., 2022).

5.2.4 Research Gaps & Methodology

Here, we discuss the main differences between the 3D-SCPTOP model proposal and those reviewed by specifically focusing on procurement transport. The key distinction of the 3D-SCPTOP model compared to the existing literature is the incorporation of 3D constraints and vertical stability constraints to model transport capacity with a view to providing more realistic solutions and reducing the likelihood of infeasible outcomes. Table 7.1 presents a comparison of inventory modelling, transport modelling, and supply chain configuration between the 3D-SCPTOP model proposal and those currently in the literature. Hence, the 3D-SCPTOP approach is based on a SS supply chain configuration, and takes into account the FTL transportation mode.

Article	SC Configuration	Decision level	Transportation Mode	Transportation Capacity Constraint	Transportation Cost	Inventory Capacity Restriction	RTIs	Inventory Costs	Safety Stocks	Modelling Framework
(Guo et al., 2024)	MM	TO	FTL	UH	X	X	S|NF	X		MINLP
(Berman & Wang, 2006)	MM	T	FTL|LTL	V	X			X		MINLP
(Sancak & Salman, 2011)		O	LTL|FTL	V	X			X	X	MILP
(Schöneberg et al., 2010)	MS	T	FTL|LTL	V	X	X		X		MILP
(Schöneberg et al., 2013)	MS	O	FTL|LTL	V	X	X		X		MILP
(Díaz-Madroñero, Mula, & Peidro, 2014)	SS	O	FTL	UH	X			X	X	FMOILP
(Hein & Almeder, 2016)	MS	O	MR	V	X			X		MILP
(Díaz-Madroñero et al., 2017)	MS	T	FTL|LTL|MR	V	X	X		X	X	MILP
(Meyer & Amberg, 2018)	MS	T|O	MR|FTL	V	X	X		X		MILP
(Chitsaz et al., 2019)	MS	O|T	MR	V	X	X		X		MILP
(Baller et al., 2022)	MS	O	MR|FTL|LTL|CES	V|W	X	X			X	MILP
(Hashemi-Amiri et al., 2023)	MS	T	MR	V	X					CCP
3D-SCPTOP	SS	O	FTL	3D-Constraints	X	X		X	X	MILP
(Y. Zhang et al., 2024)	MS	TO	MR	UH	X	X	M|NF	X		ILP

Abbreviations. **T** (tactical), **O** Operational; **FTL** (full-truck load), **LTL** (less-than-truck load), **MR** (milk-run), **CES** (courier express service); **V** (volumetric), **W** (weight), **UH** Unit Holding; **CCP** chance-constrained programming

Table 5.2: Comparison of procurement transport models

Article	Problem type	Resolution procedure	Orientation	Weight	Load Balance	Stability	Time-Depenent
(Zhu & Lim, 2012)	SLOP|SKP	TSM	X	X			
(Gonçalves & Resende, 2013)	SBSBPP	BRKGA	X	X		X	
(Paquay et al., 2016)	SBSBPP	EM	X	X	X	X	
(Mahvash et al., 2018)	SBSBPP	CG	X	X			
(Alonso et al., 2020)	SSSCSP	GRASP	X	X	X	X	X
(Tresca et al., 2022)	SBSBPP	MH	X	X	X	X	
(Harrath, 2022)	SBSBPP	H	X	X	X		
(Li et al., 2022)	SBSBPP|MSSCSP	HALNs	X	X		X	
(M. Chen et al., 2023)	BSDPP	HBBO	X	X			
3D-SCPTOP	SSSCSP & MILOPP	MH	X	X	X	X	X

Abbreviations. **T**SM (Tree search method), **BRKGA** Biased Randomized Jey Genetic Algorithm; **EM** (Exact Metods), **MH** (MatHeuristic), **CG** (Column Generation), **H** (Heuristic); **HCLP** (Hybrid Adaptive Large Neighborhood Search), **HBBO** (Hybrid Biogeography-Based Optimization)

Table 5.3: Comparison of C&P problems

On the other hand, the 3D-SCTOP model diverges from those proposed in the packaging literature in several aspects. Firstly, the loading problem takes into account different time periods. Therefore, the specific container (small items or boxes in the $C\&P$ literature) that we utilise are yet to be defined, as the choice of container depends on the inventory levels of each product. Also, the objective function depends on inventory and transport costs, which are inherently connected to the loading problem at hand. Thefore, we simultaneously solve an input minimisation and output maximisation problem. Both of these are a result of transport minimisation and inventory costs. Input minimisation comes because we want to find the minimum number of trucks needed. Meanwhile, output maximization comes from the FTL and the transport costs.

Based on the literature review and the identified research gaps, our proposal is structured according to the steps depicted in Figure 5.1.

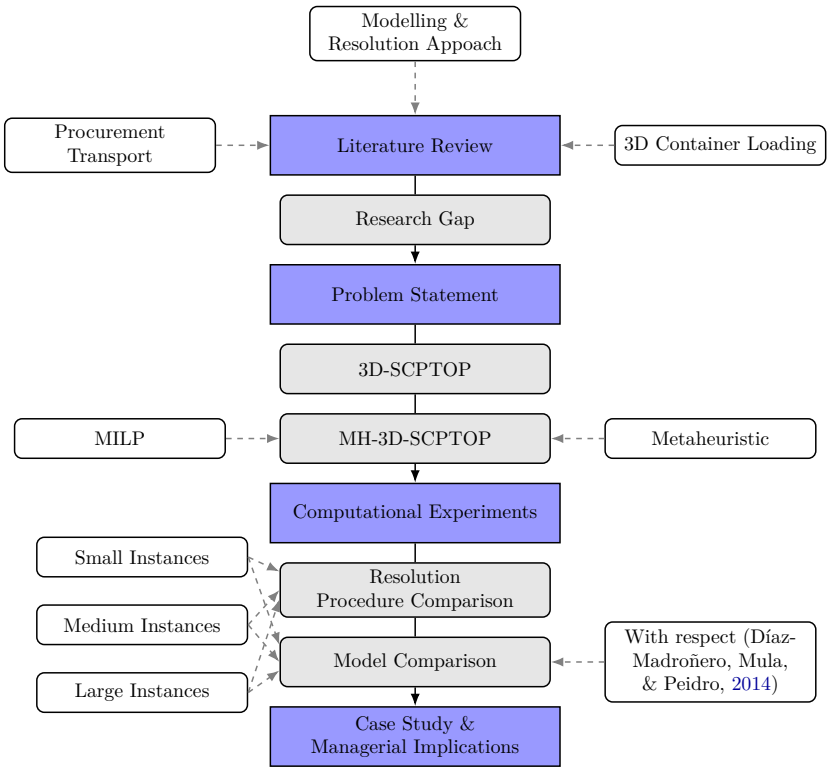


Figure 5.1: Research Methodology

5.3 Problem Description

The main objective of the presented problem is to minimise the transportation and inventory holding costs of a manufacturing site by integrating procurement transport, RTI management and inventory decisions with 3D full-truck loading constraints. The supply chain under study has a single source and a single destination. FTL is employed as the mode of transport. Products are shipped in different RTI types. RTIs are transported inside trucks, and the loading configuration is determined. Therefore, a distinction is made to the classic nomenclature of $C\&P$ problems. Hereafter, small objects are RTIs and large objects are trucks.

The assembler's demand drives the procurement process because it determines the optimal quantity of the products to procure from the supplier to the manufacturing site. Raw materials and assembled parts are produced at the supplier following the production plan. Then, they are packed in their corresponding RTI. The full RTIs with RM and AP are transported using an FTL mode of transport with 3D loading constraints. Each RTI can only carry the same product type at a given time. Furthermore, each RM or AP can only be shipped in one RTI type. Once the full RTIs are at the plant, the RM and AP are used for production, and the RTIs are emptied. Once emptied, they are folded and shipped back

to the supplier using FTL transport with 3D loading constraints. Figure 5.2 is an example of the assignment of RM and AP to RTI types, where three container types and six product types are represented in different colours. The arrow indicates the amount of product that can be shipped in the containers.

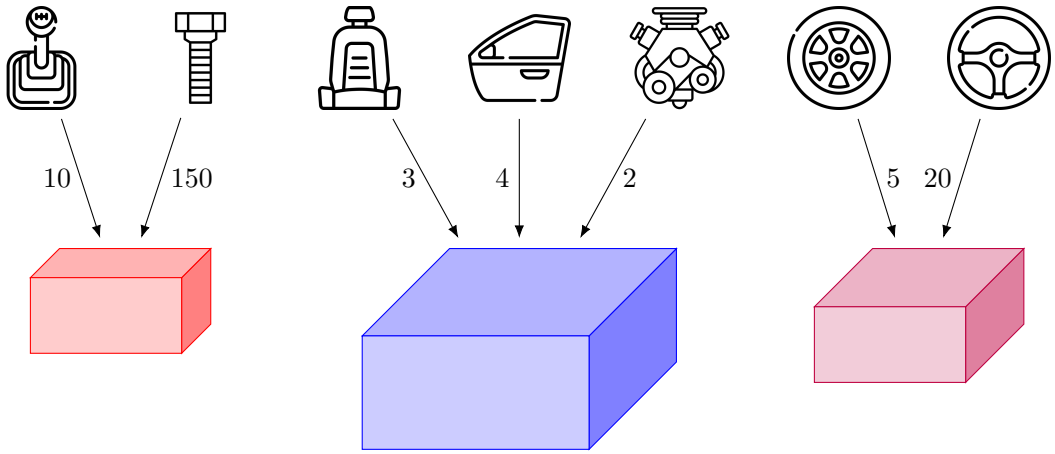


Figure 5.2: Product to type of reusable container

In our problem, two distinct transport decisions are made: procuring RM and AP for the manufacturing plant with full RTIs. And the supply of empty RTIs to the supplier.

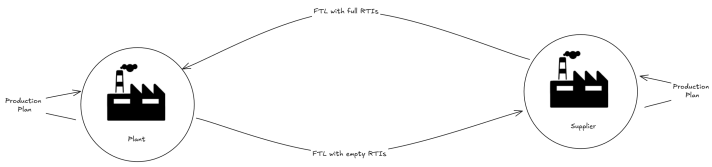


Figure 5.3: Enter Caption

Several constraints and assumptions are considered in this model formulation. Firstly, the physical transport constraints ensure that RTIs do not overlap and are contained in the truck. A truck's maximum weight limit must not be exceeded. Maintaining the static stability of containers during transport is crucial to prevent tipping or shifting because it could damage RM and AP. Rotations of containers about z-axes are allowed. As FTL is the mode of transport employed, trucks should be loaded to achieve a minimum utilisation percentage. Furthermore, we assume that the supplier's production plans and the plan are synchronised to guarantee the production of the manufacturing site. Minimum inventory coverage constraints are considered, subject to the product's future demand. Thus, the problem can be formulated in the following way:

Initial data:

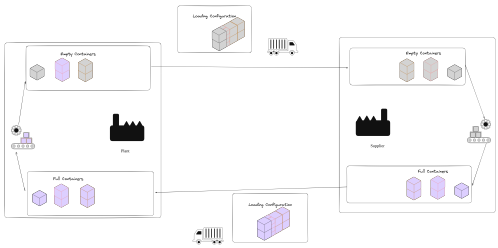


Figure 5.4: Enter Caption

- Product data: initial inventory, demand for each time period of the plant in terms of full RTIs, the type of reusable container in which it must travel, production demand for each time period at the supplier in terms of RTIs, and the weight of the RTI when loaded with the given product
- Reusable container data: RTI dimensions when full and folded.
- Transport data: number of available trucks, minimum utilisation percentage, dimensions and maximum weight

To determine:

- The number of trucks used daily with their corresponding loading configuration.
- The inventory level in terms of full RTIs for each product in the factory and supplier, given that everything goes according to plan.
- The inventory level of folded RTIs in the factory and supplier.

Subject to:

- Loading restrictions (i,e: containers cannot overlap, rotations about z axes are allowed, the maximum weight of a truck can not be exceeded, static stability)
- Minimum coverage stock

In order to:

- Minimise the total transportation cost
- Minimise the total inventory holding cost

5.4 Mathematical model

The 3D-SCPTOP mathematical programming model integrates container loading constraints and procurement transport decisions. It differs from traditional 3D container packing problems in two significant aspects. Firstly, the quantity of containers required for loading depends on product demand and the inventory levels of the manufacturing site. Therefore, the specific products to be transported are identified to minimise transport and inventory costs. As inventory levels depend on past loading decisions, a time dimension has to be taken into account. Secondly, it combines output maximisation problems and input minimisation. On the one hand, the number of trucks to be used is minimised, i.e. input minimisation. On the other hand, trucks must be fully loaded, i.e. output maximisation. It also extends

traditional procurement transport planning approaches by taking into account 3D loading constraints and RTI management.

Table 7.2 presents the nomenclature of the 3D-SCPTOP model.

Table 5.4: Nomenclature of 3D-SCPTOP

Indices	
R	Set of RTI types ($r = 1, \ldots, R$)
I	Set of RTIs ($i = 1, \ldots, I$)
K	Set of trucks ($k = 1, \ldots, K$)
N	Set of products ($n = 1, \ldots, N$)
L	Set of loading configurations ($l = 1, 2, \ldots, L$)
T	Set of planning periods *(t =1, ... T)*
G	Set corresponding to the supplier and plant nodes $G = \{s, p\}$
Set of indices	
$M(r)$	Set of RM or AP that are carried in RTI $r \in R$, $n = 1, \ldots, M(r)$
Parameters	
cK	Cost of using a truck
lt	Lead transport time between the plant and the supplier and viceversa.
i_{gn}	Initial inventory level of $n \in N$ in $g \in G$ in terms of full RTIs
ie_{gr}	Initial inventory level of empty RTIs $r \in R$ in $g \in G$ RTIs
ic_{gn}	Inventory cost of $n \in N$ at $g \in G$ in terms of full RTIs
d_{gnt}	Production plan of $n \in N$ at $g \in G$ during time period $t \in T$
q_{rt}	Initial quantity of empty RTIs $r \in R$ arriving at supplier $s \in G$ in time period $t \in (1, \ldots, lt)$
q_{nt}	Initial quantity of full RTIs of $n \in R$ arriving at p in time period $t \in (1, \ldots, lt)$
lc_{lrg}	Number of containers of type c used in loading configuration l arriving at $g \in G$
ss_{gnt}	Security stock for product $n \in N$ in time period $t \in T$ in $g \in G$ in terms of full RTIs
ss_{gtr}	Security stock of empty RTIs $r \in R$ in time period $t \in T$ in $g \in G$
w_n	Weight of a container carrying product n
pK	Maximum weight capacity of a truck
pK	Maximum weight capacity of truck a truck
Γ_c	Volume of a container type c
Γ	Volume of a truck
α	Minimum volume utilisation of a truck
M	A large enough number to make the logic of the model work;
p_{gr}, q_{gr}, r_{gr}	length, width, and height of a container of type c;
pK, qK, rK	Length, width and height of a truck;
Decision Variables	
I_{gnt}	Inventory of product $n \in N$ at the end of $t \in T$ time period at $g \in G$ in terms of full RTIs
Ie_{gtr}	Inventory of empty RTIs $r \in R$ at the end of $t \in T$ time period at $g \in G$
Y_{gkt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ arriving at $g \in G$ and zero otherwise
Y_{gklt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ departing from $g \in G$ using loading configuration l, 0 otherwise

Q_{nt}	Integer variable indicating the number of RM or AP $n \in N$ in transport during time period t to the manufacturing plant p
Q_{rt}	Integer variable the number of empty RTIs $r \in R$ in transport during time period t to the supplier s
V_{gklrt}	Number of RTIs $r \in R$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$
V_{kglnt}	Number of full RTIs with RM or AP $n \in N$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$
I_{nt}	Inventory of product n at the end of time period t
X_{gil}	Binary variable indicating whether container i is used in loading configuration l
X_{gilr}	Binary variable indicating whether container i is going to be of type c for loading configuration l
V_{gklrt}	Number of containers of type c in truck k using loading configuration l at time period t
V_{kglnt}	Integer variable indicating the quantity of containers carrying product n supplied on truck k using loading configuration l at time period t.
PX_{gil}, PY_{gil}, PZ_{gil}	Coordinates of the lower left front corner of container i on the truck k during planning period t;
LX_{gilr}, LY_{gilr} , LZ_{gilr}	Defines whether the length of container i of type c is on the same direction as the x-, y- or z-axis in loading configuration l. For example, LX_{gilr} equals one if the length of container i of type c is is on the same direction as the x-axis, otherwise LX_{gilr} equals 0;
WX_{gilr} , WY_{gilr} , WZ_{gilr}	Defines whether the width of container i of type c is on the same direction as the x-, y- or z-axis on truck k axis during planning period t. For example, WX_{gilr} equals 1 if the width of container i of type c is is on the same direction as the x-axis, otherwise WX_{gilr} equals 0;
HX_{gilr}, HY_{gilr} , HZ_{gilr}	Defines whether the height of container i of type c is on the same direction as the x-, y- or z-axis on the truck k axis during planning period t. For example, HX_{gilr} equals 1 if the width of container i of type c is is on the same direction as the x-axis, otherwise HX_{gilr} equals 0
A_{gijl}, B_{gijl}, C_{gijl}, D_{gijl}, E_{gijl}, F_{gijl}	Binary variables indicating the relative position between two boxes. Variable A_{gijl} equals 1 if container i is to the left of container j on truck k during planning period t. Similarly, variables B_{gijl},C_{gijl}, D_{gijl} , E_{gijl} , F_{gijl} indicate whether container i is to the right, behind, in front, on top or below container j, respectively, on truck k during planning period t. These variables are only necessary when $i \neq k$.

We wish to minimise the total cost, which is given by transportation and inventory costs.

$$Min\ z = \sum_{g \in G} \sum_{k \in K} \sum_{t \in T} Y_{gkt} \cdot cK + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} I_{gnt} \cdot ic_{gn} \tag{5.1}$$

(5.60) - (5.62) define the inventory level of RM or AP according to the production demand

and supply of RM or AP from the supplier s at the plant p

$$I_{pnt} = I_{pnt'} - d_{pnt} + Q_{nt'} \qquad \forall n \in N, t \in T : t \geqslant lt + 1, t' := t - lt \tag{5.2}$$

$$I_{pnt} = I_{pnt'} - d_{pnt} + q_{nt} \qquad \forall n \in N, t \in (2, \ldots, lt) :, t' := t - lt \tag{5.3}$$

$$I_{pnt} = i_{pn} - d_{pnt} + q_{nt} \qquad \forall n \in N : t := 1 \tag{5.4}$$

$$Ie_{ptr} = Ie_{pt'r} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t \in T : t \geqslant 2 \tag{5.5}$$

$$Ie_{ptr} = ie_{pr} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t := 1 \tag{5.6}$$

(5.65) - (5.66) the empty RTIs according to the production demand and supply of RTIs from the plant p at the supplier s

$$I_{snt} = I_{snt'} + d_{snt} - Q_{nt} \qquad \forall n \in N, t \in T, t' \in T, : t \geqslant 2, t' := t - lt \tag{5.7}$$

$$I_{snt} = i_{sn} + d_{snt} - Q_{nt} \qquad \forall r \in R : t := 1 \tag{5.8}$$

$$Ie_{str} = Ie_{st'r} + Q_{rt'} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R, t \in T : t \geqslant lt + 1, t' := t - lt \tag{5.9}$$

$$Ie_{str} = Ie_{st'r} + q_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R, t \in (2, \ldots, lt) :, t' := t - lt \tag{5.10}$$

$$Ie_{str} = ie_{pr} + q_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R : t = 1 \tag{5.11}$$

Each product must satisfy a minimum security stock ss_{gnt}.

$$I_{gnt} \geqslant ss_{gnt} \quad \forall g \in G, n \in N, t \in T \tag{5.12}$$

(5.71) imposes that a truck k can only use one loading configuration l with direction $g \in G$ in time period t.

$$\sum_{l \in L} Y_{gklt} = Y_{gkt} \quad \forall g \in G, k \in K, t \in T \tag{5.13}$$

$$X_{gil} = \sum_{r \in R} X_{gilr} \quad \forall g \in G, i \in I, l \in L \tag{5.14}$$

$$V_{glr} = \sum_{i \in I} X_{gilr} \quad \forall g \in G, r \in R, l \in L \tag{5.15}$$

(5.16) ensures that a truck is used if it is carrying containers. Meanwhile, (5.17) - (5.18) impose that the truck k at time period t using loading configuration l uses the same loading configuration as l.

$$V_{gklrt} \leq M(1 - Y_{gklt}) \qquad \forall g \in G, r \in R, k \in K, t \in T \tag{5.16}$$

$$V_{gklrt} + M(1 - Y_{gklt}) \geq V_{glr} \qquad \forall g \in G, r \in R, k \in K, t \in T \tag{5.17}$$

$$V_{gklrt} \leq V_{glr} + M(1 - Y_{gklt}) \qquad \forall g \in G, r \in R, k \in K, t \in T \tag{5.18}$$

Since a product can only be carried by one container type (5.74) sets that the number of containers of type c carried at truck k using loading configuration l at time period c is

equal to the sum of containers carried of products n which containers of type c can carry. Moreover, (5.75) sets the total amount of product n being transported from s to p in time period t, and (5.76) sets the total amount of RTIs r being transported from p to s in time period t,

$$V_{pklrt} = \sum_{n \in M(r)} V_{pklnt} \qquad \forall k \in K, l \in L, t \in T, r \in R \tag{5.19}$$

$$Q_{nt} = \sum_{l \in L} \sum_{k \in K} V_{kslnt} \qquad \forall n \in N, t \in T \tag{5.20}$$

$$Q_{rt} = \sum_{l \in L} \sum_{k \in K} V_{pklrt} \qquad \forall r \in R, t \in T \tag{5.21}$$

(5.77) guarantees that the maximum weight allowed for truck k is not surpassed.

$$\sum_{l \in L} \sum_{n \in N} w_n V_{kglnt} \leq pK \qquad \forall k \in K, t \in T, g \in G \tag{5.22}$$

We ensure that two containers are not overlapped by their relative positions. (5.23) and (5.24) to identify whether container i is to the left or the right of container j.

$$PX_{gil} + \sum_{r \in R} (p_{gr} LX_{gilr} + q_{gr} WX_{gilr} + r_{gr} HX_{gilr}) \leq PX_{gjl} + (1 - A_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{5.23}$$

$$PX_{gjl} + \sum_{r \in R} (p_{gr} LX_{gjlr} + q_{gr} WX_{gjlr} + r_{gr} HX_{gjlr}) \leq PX_{gil} + (1 - B_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{5.24}$$

Similarly, (5.25)-(5.28) determine whether one container is ahead, behind, on top of or below one another.

$$PY_{gil} + \sum_{r \in R} (q_{gr} WY_{gilr} + p_{gr} LY_{gilr} + r_{gr} HY_{gilr}) \leq PY_{gjl} + (1 - C_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{5.25}$$

$$PY_{gjl} + \sum_{r \in R} (q_{gr} WY_{gjlr} + p_{gr} LY_{gjlr} + r_{gr} hy_{gjlr}) \leq PY_{gil} + (1 - D_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{5.26}$$

$$PZ_{gil} + \sum_{r \in R} (r_{gr} HZ_{gilr} + q_{gr} WZ_{gilr} + p_{gr} LZ_{gilr}) \leq PZ_{gjl} + (1 - E_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{5.27}$$

$$PZ_{gjl} + \sum_{r \in R} (r_{gr} HZ_{gjlr} + q_{gr} WZ_{gjlr} + p_{gr} LZ_{gjlr}) \leq PZ_{gil} + (1 - F_{gijl})\, M \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{5.28}$$

(5.29) ensures that if containers i and j are being used, they must be to the left, right, in front, behind, on top of or below one another.

$$A_{gijl} + B_{gijl} + C_{gijl} + D_{gijl} + E_{gijl} + F_{gijl} + (1 - X_{gil}) + (1 - X_{gjl}) \geq 1 \qquad \forall g \in G, i \in I, j \in I : i \neq j, t \in T \tag{5.29}$$

(5.30) links the container type chosen with orientation.

$$LX_{gilr} + LY_{gilr} + LZ_{gilr} + WX_{gilr} + WY_{gilr} + WZ_{gilr} + HX_{gilr} + HY_{gilr} + HZ_{gilr} \leq 3 \cdot X_{gilr} \qquad \forall g \in G, r \in R, i \in \tag{5.30}$$

(5.31) - (5.33) restrict the solution so that containers are in the truck.

$$PX_{gil} + \sum_{r \in R} (p_{gr} LX_{gilr} + q_{gr} WX_{gilr} + r_{gr} HX_{gilr}) \leq p_k + (1 - X_{gil})\, M \quad \forall g \in G, i \in I, k \in K, t \in T \tag{5.31}$$

$$PY_{gil} + \sum_{r \in R} (q_{gr} WY_{gilr} + p_{gr} LY_{gilr} + r_{gr} ZY_{gilr}) \leq q_k + (1 - X_{gil})\, M \quad \forall g \in G, i \in I, k \in K, t \in T \tag{5.32}$$

$$PZ_{gil} + \sum_{r \in R} (r_{gr} HZ_{gilr} + p_{gr} LZ_{gilr} + q_{gr} WZ_{gilr}) \leq r_k + (1 - X_{gil})\, M \quad \forall g \in G, i \in I, k \in K, t \in T \tag{5.33}$$

A possible orientation of a container must satisfy that the length, width and height of container i are x-, y- or z-axis a single axis. Moreover, these have to be different from one another. Therefore, the following constraints must hold.

$$LX_{gilr} + LY_{gilr} + LZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{5.34}$$

$$WX_{gilr} + WY_{gilr} + WZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{5.35}$$

$$HX_{gilr} + HY_{gilr} + HZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{5.36}$$

$$LX_{gilr} + WX_{gilr} + HX_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{5.37}$$

$$LY_{gilr} + WY_{gilr} + HY_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{5.38}$$

$$LZ_{gilr} + WZ_{gilr} + HZ_{gilr} = X_{gilr} \qquad \forall g \in G, r \in R, i \in I, t \in T \tag{5.39}$$

To ensure trucks are loaded to minimum utilisation, the equation constrains the solutions so the truck must at least be loaded to α_k of its capacity.

$$\alpha_k \leq \frac{\sum_{c,l} V_{gklrt} \Gamma_c}{\Gamma_k} \qquad \forall g \in G, k \in K, t \in T \tag{5.40}$$

Finally, we define the domain of the variables.

$$I_{nt} \geq 0 \qquad \forall n \in N, t \in T \tag{5.41}$$

$$Y_{kt} \in \{0,1\} \qquad \forall k \in K, t \in T \tag{5.42}$$

$$X_{gil}, X_{gilr}, X^n_{gilr} \in \{0,1\} \qquad \forall r \in R, i \in I, k \in K, n \in N, t \in T \tag{5.43}$$

$$V^n_t, V^k_{int} \in \mathbb{N} \cup \{0\} \qquad \forall i \in I, k \in K, n \in N, t \in T \tag{5.44}$$

$$PX_{gil}, PY_{gil}, PY_{gil} \geq 0 \qquad \forall i \in I, t \in T, k \in K \tag{5.45}$$

$$LX_{gilr}, LY_{gilr}, LZ_{gilr}, WX_{gilr}, WY_{gilr}, WZ_{gilr}, HX_{gilr}, HY_{gilr}, HZ_{gilr} \in \{0,1\} \qquad \forall r \in R, i \in I, k \in K, t \in T \tag{5.46}$$

$$A_{gijl}, B_{gijl}, C_{gijl}, D_{gijl}, E_{gijl}, F_{gijl} \in \{0,1\} \qquad \forall i \in I, j \in J, k \in K, t \in T \tag{5.47}$$

Regarding vertical stability, the 3D-SCPTOP model is extended to ensure that loaded boxes resist the force of gravity. Similar to (Paquay et al., 2016), vertical stability is ensured by determining if the vertices of containers are supported by one another when they do not directly contact the ground. Furthermore, we impose that for an RTI to be in top of another, it has to have all four corners in top of another RTI, unless it's at the ground level. The four bottom corners are denoted by a tuple (u, u') where $u, u' \in U = \{0,1\}$. Parameter u refers to the position on the x-axis (with $u = 0$ indicating a minimum x value and $u = 1$ a maximum x value). Parameter u' is defined analogously for the y-axis (see Figure 5.5).

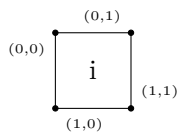


Figure 5.5: Description of the bottom corners position

The following additional variables are defined:

Variables	
T_{gijl}	Binary variable equal to 1 if RTI $i \in I$ is directly on top of $j \in I$ in loading configuration l.
GL_{gil}	Binary variable indicating if container i is on the ground level on truck k during time period t.
$O_{gijuu'l}$	Binary variable equal to 1 if the bottom corner (u, u') of container i is immediately supported by container j on truck k during time period t.

Constraints (5.48) and (5.49) link the z-coordinate of container i with the ground–level variable GL_{gil}:

$$M\Big(1 - GL_{gil}\Big) \geq PZ_{gil} \quad \forall i \in I,\ t \in T \tag{5.48}$$

$$1 - GL_{gil} \leq M\, PZ_{gil} \quad \forall i \in I,\ t \in T \tag{5.49}$$

Constraints (5.50) and (5.51) imply that for $O_{gijuu'l}$ to be active, the heights of containers i and j must be compatible:

$$PZ_{gil} \leq PZ_{gjl} + \sum_{r \in R}\Big(r_{gr}\, HZ_{gjlr} + q_{gr}\, WZ_{gjlr} + p_{gr}\, LZ_{gjlr}\Big) + (1 - O_{gijuu'l})M \quad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{5.50}$$

$$PZ_{gil} + (1 - O_{gijuu'l})M \geq PZ_{gjl} + \sum_{r \in R}\Big(r_{gr}\, HZ_{gjlr} + q_{gr}\, WZ_{gjlr} + p_{gr}\, LZ_{gjlr}\Big) \quad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{5.51}$$

Furthermore, the coordinates of the corner (u, u') of container i must lie within the XY projection of container j. Constraints (5.52) and (5.53) link the x-dimensions, and constraints (5.54) and (5.55) link the y-dimensions. Parameter ϵ assures that the corner (u, u') has at least ϵ support.

$$PX_{gil} + u\left(\sum_{r \in R}(p_{gr}\, LX_{gilr} + q_{gr}\, WX_{gilr} + r_{gr}\, HX_{gilr})\right) + M(1 - O_{gijuu'l}) \geq PX_{gjl} + \epsilon\, u \quad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{5.52}$$

$$PX_{gil} + u\left(\sum_{r \in R}(p_{gr}\, LX_{gilr} + q_{gr}\, WX_{gilr} + r_{gr}\, HX_{gilr})\right) \leq PX_{gjl} + \sum_{r \in R}(p_{gr}\, LX_{gjlr} + q_{gr}\, WX_{gjlr} + r_{gr}\, HX_{gjlr}) + M(1 - O_{gijuu'l}) - (1 - u)\epsilon \quad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{5.53}$$

$$PY_{gil} + u'\left(\sum_{r \in R}(q_{gr}\, WY_{gilr} + p_{gr}\, LY_{gilr} + r_{gr}\, HY_{gilr})\right) + M(1 - O_{gijuu'l}) \geq PY_{gjl} + \epsilon\, u' \quad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{5.54}$$

$$PY_{gil} + u'\left(\sum_{r \in R}(q_{gr}\, WY_{gilr} + p_{gr}\, LY_{gilr} + r_{gr}\, HY_{gilr})\right) \leq PY_{gjl} + \sum_{r \in R}(q_{gr}\, WY_{gjlr} + p_{gr}\, LY_{gjlr} + r_{gr}\, HY_{gjlr}) + M(1 - O_{gijuu'l}) - (1 - u')\epsilon \quad \forall i, j \in I,\ i \neq j,\ k \in K,\ (u, u') \in U \times U,\ t \in T \tag{5.55}$$

Constraint (5.56) ensures that either a used container is at the bottom of the truck or all four bottom corners are supported by other boxes:

$$\sum_{j \in I,\ (u,u') \in U \times U} O_{gijuu'l} + M\Big(GL_{gil} + 1 - X_{gil}\Big) \geq 4 \quad \forall i \in I,\ k \in K,\ t \in T \tag{5.56}$$

Constraint (5.57) limits $O_{gijuu'l}$ to be active only if container j is used during planning period t:

$$O_{gijuu'l} \geq X_{gjl} \quad \forall g \in G,\ i,j \in I,\ (u,u') \in U \times U,\ k \in K,\ t \in T \tag{5.57}$$

Additionally, to allow only 90 degree rotations along the z-axis the following constraint is added:

$$HZ_{gilr} \leq X_{gilr} \quad \forall g \in G,\ r \in R,\ i \in I,\ k \in K,\ t \in T \tag{5.58}$$

Finally, the following equations link the RTI stacking variables:

$$4\,T_{gijl} = \sum_{(u,u') \in U \times U} O_{gijuu'l} \quad \forall g \in G,\ i,j \in I,\ l \in L,\ i \neq j$$

$$\sum_{j \in I} T_{gijl} \leq 1 \quad \forall g \in G,\ i \in I,\ l \in L$$

5.5 Matheuristic Procedure

As previously mentioned, exact methods provided by MILP solvers fall short for tackling real world instances with the current computation technology. Therefore, a matheuristic approach is proposed, Math-3D-SCPTOP. It is based on decoupling the two sub-processes involved in the problem: finding 3D loading configurations of RTIs inside the trucks and determining the number of trucks and what to carry in each time period in order to minimize inventory and transport costs.

This matheuristic is based on two main components. First a MILP model is defined to determine which loading configurations from a given set should be used in order to minimise the transport and inventory costs. Second a metaheuristic approach is used to identify these loading configurations.

5.5.1 PLC-SCPTOP

The following MILP model, dubbed as PLC-SCPTOP, uses predefined loading configurations of container types to determine which loading configurations to use, the products to carry and the number of trucks in order to minimise inventory and transport costs. The predefined loading configurations must already obey the spatial constraints defined in 3D-SCPTOP. Theoretically, if every conceivable loading configuration were incorporated into the model, the optimal solution of PLC-SCPTOP would align with that of 3D-SCPTOP. Enumerating all of the possible loading configurations of a 3D Bin Packing Problems would take an exponential number of memory space and cpu time with respect the number of container types. Moreover, solving the PLC-SCPTOP would become even more time-consuming, assuming the MILP program could be fully allocated in memory.

Table 5.5

Table 5.5: Nomenclature of Math-3D-SCTOP

Indices	
R	Set of RTI types ($r = 1, \ldots, R$)
K	Set of trucks ($k = 1, 2, \ldots, K$)
N	Set of products *(n = 1,. . . ,N)*

L	Set of loading configurations ($l = 1, 2, \ldots, L$)
T	Set of planning periods *(t =1, 2... T)*
G	Set corresponding to the supplier and plant nodes $G = \{s, p\}$
Set of indices	
$M(r)$	Set of RM or AP that are carried in RTI $r \in R$, $n = 1, \ldots, M(r)$
Parameters	
cK	Cost of using a truck
lt	Lead transport time between the plant and the supplier and viceversa.
i_{gn}	Initial inventory level of $n \in N$ in $g \in G$ in terms of full RTIs
ie_{gr}	Initial inventory level of empty RTIs $r \in R$ in $g \in G$ RTIs
ic_{gn}	Inventory cost of $n \in N$ at $g \in G$ in terms of full RTIs
d_{gnt}	Production plan of $n \in N$ at $g \in G$ during time period $t \in T$
q_{rt}	Initial quantity of empty RTIs $r \in R$ arriving at supplier $s \in G$ in time period $t \in (1, \ldots, lt)$
q_{nt}	Initial quantity of full RTIs of $n \in R$ arriving at p in time period $t \in (1, \ldots, lt)$
lc_{lrg}	Number of containers of type c used in loading configuration l arriving at $g \in G$
ss_{gnt}	Security stock for product $n \in N$ in time period $t \in T$ in $g \in G$ in terms of full RTIs
ss_{gtr}	Security stock of empty RTIs $r \in R$ in time period $t \in T$ in $g \in G$
w_n	Weight of a container carrying product n
k_t	Numbers of trucks available at time period t
pK	Maximum weight capacity of a truck
Decision Variables	
I_{gnt}	Inventory of product $n \in N$ at the end of $t \in T$ time period at $g \in G$ in terms of full RTIs
Ie_{gtr}	Inventory of empty RTIs $r \in R$ at the end of $t \in T$ time period at $g \in G$
Y_{gkt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ arriving at $g \in G$ and zero otherwise
Y_{gklt}	Binary variable, if one then truck $k \in K$ is used in time period $t \in T$ departing from $g \in G$ using loading configuration l, 0 otherwise
Q_{nt}	Integer variable indicating the number of RM or AP $n \in N$ in transport during time period t to the manufacturing plant p
Q_{rt}	Integer variable the number of empty RTIs $r \in R$ in transport during time period t to the supplier s
V_{gklrt}	Number of RTIs $r \in R$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$
V_{kglnt}	Number of full RTIs with RM or AP $n \in N$ in truck $k \in K$ using loading configuration $l \in L$ at time period $t \in T$ departing from $g \in G$

We wish to minimise the total cost, which is given by transportation and inventory costs.

$$Min\ z = \sum_{g \in G} \sum_{k \in K} \sum_{t \in T} Y_{gkt} \cdot cK + \sum_{g \in G} \sum_{n \in N} \sum_{t \in T} I_{gnt} \cdot ic_{gn} \tag{5.59}$$

(5.60) - (5.62) define the inventory level of RM or AP according to the production demand

and supply of RM or AP from the supplier s at the plant p

$$I_{pnt} = I_{pnt'} - d_{pnt} + Q_{nt'} \qquad \forall n \in N, t \in T : t \geqslant lt + 1, t' := t - lt \tag{5.60}$$
$$I_{pnt} = I_{pnt'} - d_{pnt} + q_{nt} \qquad \forall n \in N, t \in (2, \ldots, lt) :, t' := t - lt \tag{5.61}$$
$$I_{pnt} = i_{pn} - d_{pnt} + q_{nt} \qquad \forall n \in N : t := 1 \tag{5.62}$$
$$Ie_{ptr} = Ie_{pt'r} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t \in T : t \geqslant 2 \tag{5.63}$$
$$Ie_{ptr} = ie_{pr} + \sum_{n \in M(r)} d_{pnt} - Q_{rt} \qquad \forall r \in R, t := 1 \tag{5.64}$$

(5.65) - (5.66) the empty RTIs according to the production demand and supply of RTIs from the plant p at the supplier s

$$I_{snt} = I_{snt'} + d_{snt} - Q_{nt} \qquad \forall n \in N, t \in T, t' \in T, : t \geqslant 2, t' := t - lt \tag{5.65}$$
$$I_{snt} = i_{sn} + d_{snt} - Q_{nt} \qquad \forall r \in R : t := 1 \tag{5.66}$$
$$Ie_{str} = Ie_{st'r} + Q_{rt'} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R, t \in T : t \geqslant lt + 1, t' := t - lt \tag{5.67}$$
$$Ie_{str} = Ie_{st'r} + q_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R, t \in (2, \ldots, lt) :, t' := t - lt \tag{5.68}$$
$$Ie_{str} = ie_{pr} + q_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R : t = 1 \tag{5.69}$$

Each product must satisfy a minimum security stock ss_{gnt}.

$$I_{gnt} \geqslant ss_{gnt} \quad \forall g \in G, n \in N, t \in T \tag{5.70}$$

(5.71) imposes that a truck k can only use one loading configuration l with direction $g \in G$ in time period t.

$$\sum_{l \in L} Y_{gklt} = Y_{gkt} \quad \forall k \in K, g \in G, t \in T \tag{5.71}$$

(5.72) - (5.73) impose that the truck k at time period t using loading configuration l uses the same loading configuration as l.

$$V_{gklrt} + M \cdot (1 - Y_{gklt}) \geqslant lc_{lrg} \qquad \forall k \in K, g \in G, l \in L, t \in T \tag{5.72}$$
$$V_{gklrt} \leqslant Y_{gklt} \cdot lc_{lrg} + M \cdot (1 - Y_{gklt}) \qquad \forall k \in K, l \in L, g \in G, t \in T \tag{5.73}$$

Since a product can only be carried by one container type (5.74) sets that the number of containers of type c carried at truck k using loading configuration l at time period c is equal to the sum of containers carried of products n which containers of type c can carry. Moreover, (5.75) sets the total amount of product n being transported from s to p in time period t, and (5.76) sets the total amount of RTIs r being transported from p to s in time

period t,

$$V_{pklrt} = \sum_{n \in M(r)} V_{pklnt} \qquad \forall k \in K, l \in L, t \in T, r \in R \tag{5.74}$$

$$Q_{nt} = \sum_{l \in L} \sum_{k \in K} V_{kslnt} \qquad \forall n \in N, t \in T \tag{5.75}$$

$$Q_{rt} = \sum_{l \in L} \sum_{k \in K} V_{pklrt} \qquad \forall r \in R, t \in T \tag{5.76}$$

(5.77) guarantees that the maximum weight allowed for truck k is not surpassed.

$$\sum_{l,n} w_n V_{kglnt} \leq pK \qquad \forall k \in K, t \in T, g \in G \tag{5.77}$$

$$\begin{aligned}
I_{gnt} &\in \mathbb{Z}_{\geq 0} && \forall g \in G,\ n \in N,\ t \in T, \\
Ie_{gtr} &\in \mathbb{Z}_{\geq 0} && \forall g \in G,\ t \in T,\ r \in R, \\
Y_{gkt} &\in \{0,1\} && \forall k \in K,\ g \in G,\ t \in T, \\
Y_{gklt} &\in \{0,1\} && \forall k \in K,\ g \in G,\ l \in L,\ t \in T, \\
Q_{nt} &\in \mathbb{Z}_{\geq 0} && \forall n \in N,\ t \in T, \\
Q_{rt} &\in \mathbb{Z}_{\geq 0} && \forall r \in R,\ t \in T, \\
V_{gklrt} &\in \mathbb{Z}_{\geq 0} && \forall g \in G,\ k \in K,\ l \in L,\ r \in R,\ t \in T, \\
V_{kglnt} &\in \mathbb{Z}_{\geq 0} && \forall k \in K,\ g \in G,\ l \in L,\ n \in N,\ t \in T.
\end{aligned}$$

5.5.2 Meta-Heuristic Procedure

The objective of the metaheuristic is to obtain good loading configurations so the PLC-SCPTOP model can optimise transport and inventory costs while managing RTI flow. Therefore, the metaheuristic should consider the temporal nature of the problem. That is, the loading problem must consider the future demand of products and the current state of inventories.

To simulate the production and transportation, we define two separate processes. On the one hand, we have the plant processes and on the other we have the supplier processes. In each node, we first receive the RTIs, then we produce, and after that we make the transport decisions. 5.6 illustrates the both processes and the connections between them.

In both cases, we need to determine transport decisions. That is, the loading configurations and what is going to be transported. We can only transport what we already have in stock. Furhtermore, RTIs are loaded into the truck as stacks. Each RTI has a set of compatible RTIs which can be stacked with. We use a column generation technique to find the RTIs stacks that can be loaded. Then, we find a loading configuration through a 2D-SLOPP formulation, where the stacks that contain RTIs needed in the other supply chain node for the next products are prioritized through the objective function. Once the RTI loading configuration is determined, we decide RM or AP procurement for the manufacturing site, prioritizing RMs or APs which have less stock at the manufacturing site.

We will use the previously found loading configurations to consider the time dependency of the loading problem at hand. For each period $t \in T$, we will solve the PLC-SCPTOP

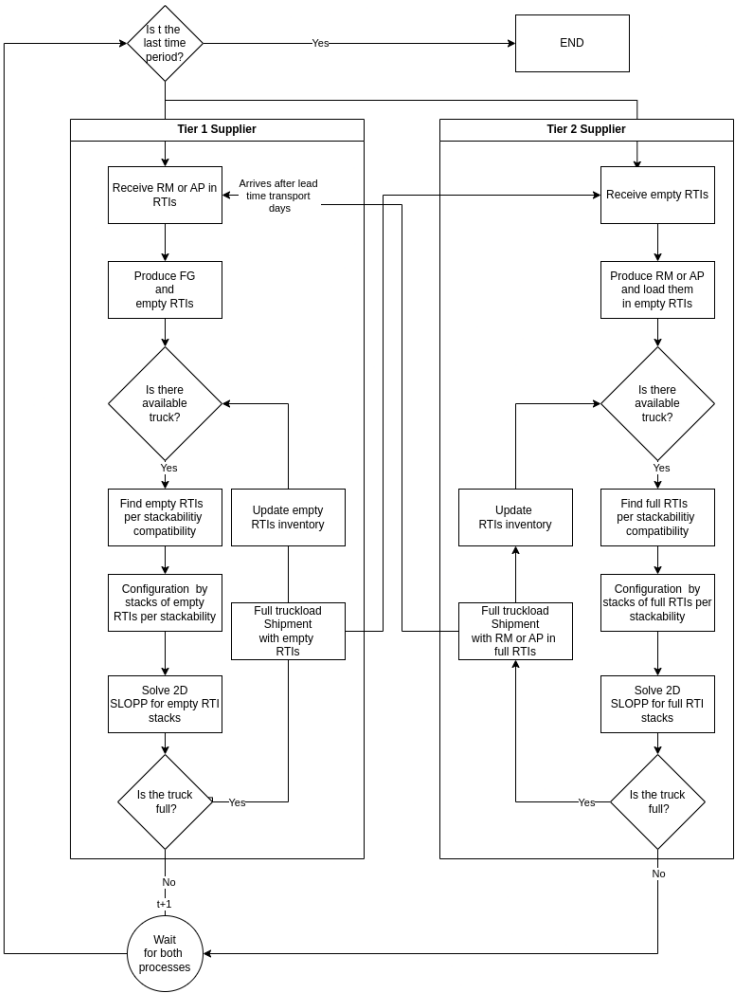


Figure 5.6: Procceses of plant and supplier

problem from day 1 to t with the current loading configurations obtained to find the inventory levels of the products at day t. Then, we find the necessary containers for the following $\delta \in \Delta$ days, where Δ is a user-defined list of parameters.

Finally, we feed the loading configurations found by Algorithms 1 and 2 to the PLC-SCPTOP mathematical modelling as parameters to find reasonable quality solutions to 3D-SCPTOP problem.

Algorithm 1 Loading Configurations

```
procedure LoadingConfigurations
    T # List of time periods
    LC=[] # List of loading configurations
    for t ∈ T do
        for δ ∈ Δ do
            IncomingTransport(t)
            Produce(t)
            while TransportResourceAvailable do
                C = ContainersToLoad(t, δ, LC)


                LC.append(PackContainers(C,t))
            end while
        end for
    end for
    LC=EliminateDuplicateOrDominatedLoadingConfigurations(LC)
    return LC
end procedure
```

Algorithm 2 Final Loading Configurations

```
procedure Final Loading Configurations(LC)
    TimePeriods # List of time spans to calculate the number of needed prod-
ucts
    TimeSpan # List of time spans to calculate the number of needed products
    for t in TimePeriods do
        for δ in TimeSpan do
            I=SolvePLC-SCPTOP(LC,t)
            C= ContainersToLoad(I,t, δ, LC)
            LC.append(PackContainers(C,t))
        end for
    end for
    LC=EliminateDuplicateOrDominatedLoadingConfigurations(LC)
    return LC
end procedure
```

Loading Procedure

To address the loading problem, will divide the problem in two parts. First, we will find through a column generation technique for the cutting stock problem to find different stacks that will be loaded. Once these stacks are found, we will

Algorithm 3 MathH-3D-SCPTOP

1: Δ_1, Δ_2 # List of time spans to calculate the number of needed products
2: LC = InitialLoadingConfiguration(Δ_1)
3: LC = FinalLoadingConfiguration(LC, Δ_2)
4: SolvePLCSCPTOP(LC,|T|))

5.6 Computational and comparative analysis

Here the 3D-SCPTOP model is validated and compared to the work by (Díaz-Madroñero, Mula, & Peidro, 2014). This fuzzy optimisation model integrates procurement transport and inventory holding decisions by contemplating transport capacity restrictions by a maximum number of containers per truck, which we refer to as SCPTOP from this point onwards. To compare the SCPTOP model, we make three key modifications: fuzzy parameters are not considered; transport capacity restriction takes into account the volume of individual containers and the truck's capacity; products can be loaded in more than one container type. The SCPTOP model is extended and defined in Appendix A. We evaluate and compare both the 3D-SCPTOP and SCPTOP models, in terms of solution quality, transport costs, inventory costs and computational times. The 3D-SCPTOP and SCPTOP mathematical formulations are solved for all the instance sets with CP-SAT from OR-Tools and Gurobi using their default settings. Additionally, as ($C\&P$) problems are NP-hard, the Meta-3D-SCPTOP algorithm has been developed to find igh-quality solutions for large real-world instances, based on a case study. So the performance of the 3D-SCPTOP, the Meta-3D-SCPTOP and the SCPTOP models is compared and evaluated as follows.

Twenty-four instances are artificially generated by a stochastic process, whose key characteristics are given as follows:

- **Demand Generation**: the demand for each product is randomly generated by deviating from a given mean by plus or minus ten percent. This introduces variability into the demand values for different products
- **Time Periods**: Half of the instances have a total of five time periods, while the other half have a total of ten time periods. This allows evaluating the model's performance over different time horizons.
- **Container Types**: Each instance includes a varying number container types, specifically two, three or four. The distribution of container types is the same among instances, which implies that an equal number of instances has the same number of containers
- **Container Dimensions**: The dimensions of each container type are randomly generated. The width, length and height of boxes are fractions of the width, length, or height of trucks. This randomness ensures diversity in the sizes of containers across the generated instances. An additional constraint is applied to the generation of containers. The volume of each container must be at least 1/n of the truck's volume, where n can take the values of four, six or eight. This ensures that containers are appropriately sized in relation to trucks

By incorporating these stochastic elements into the generation process, the 24 instances provide a diverse set of scenarios for evaluating the model's performance. Variability in

demand, time periods, box types and box dimensions introduces realistic and challenging conditions that the model needs to address. Furthermore, we use the number of time periods, the number of box types, the number of products and the maximum amount of containers needed to load a truck (in volume terms) as explanatory variables of various metrics from the models, such as total cost, transport cost, inventory cost, solution gap and solution time. The instance name is in the form of $I_n_t_c_v$, where n is product type, t is the number of time periods, c is the number of container types and v is the minimum volume of containers.

Each test is limited to 4 hours (14,400 seconds) of run-time per instance. Tests are performed on a Microsoft Windows Server 2022 Standard n with an AMD EPYC 7402 24-Core Processor running at 2,800 MHz, with 256 GB of RAM using Python as the programming language.

5.7 Practical application

This section presents a practical procurement transport application. The proposed resolution framework was evaluated using real data from an automotive industry supply chain. We validate and compare the 3D-SCPTOP solution procedures to the company's procedure, which employs predefined loading configurations that indicate the number of each container type to be carried. Finding these loading configurations is a very tedious and time-consuming manual task that is based on a spreadsheet procedure. A mathematical programming model, dubbed as PLC-SCPTOP, which replicates the current's company procedure, is then utilised to minimise the total cost, with a summation of procurement transport and inventory holding costs, and one based on these predetermined configurations.

The real-world data derived from the company's operations act as the empirical foundation for examining both the PLC-SCPTOP and the Meta-3D-SCPTOP. To broaden the scope of this analysis, we scale and manipulate these data, guided by a stochastic process, to simulate several potential operational scenarios. Setting our analysis to mirror the dynamic and unpredictable realities of supply chain operations thus enhances the validity of our evaluation and the practical relevance of the suggested metaheuristic. For this purpose, we propose three different scenarios for the created instances: normal demand, low variability demand, high variability demand. Table **??**, found in the Appendix Aoutlines the models' performance across these instances. Table 5.6 summarises models' performance and resolution times.

Table 5.6: Summary of performance for the practical application

Model	Objective Value	Best Bound	Duration (seconds)
MathH-3D-SCTOP	252235.198934	247307.820849	740.000000
Meta-3D-SCTOP	256715.145777	NaN	677.555556
PLC-SCTOP	275226.680415	271798.477640	44.670370

Figures **??** - **??** describe the performance of both the models and solution procedures in the different scenarios. The x-axis shows the scenario, while the y-axis depicts the objective value, inventory costs, or transport costs. Each colour represents a different model applied to obtain the solution.

The results reveal that an effective systematic method is proposed for solving procurement transport operational planning with 3D full-truck loading constraints. This solution is

both automated and executed within a reasonable computational timeframe. Our methodology proves particularly beneficial when there is significant discrepancy between the company's proposed loading configurations and the actual demand concerning container types. In such instances, the company is usually left with two choices: either use the fixed loading configurations and, thus, incur higher inventory and transport costs, or manually resolve the loading configuration via a spreadsheet procedure, which might lead to errors. However, one potential disadvantage of the proposed Meta-3D-SCPTOP is its performance under normal demand conditions because the demand for different container types is consistent with the loading configurations, but the metaheuristic may result in fewer optimal inventory holding costs compared to the company's current procedure. This is because the Meta-3D-SCPTOP obtains decisions locally, while exact methods are guaranteed, given enough time, to find the optimal solution.

This issue is, however, solved when utilising MathH-3D-SCPTOP because it effectively combines both heuristic and exact methods. While the heuristic provides a fast and reasonably good solution for the initial loading configurations, the exact method improves it by further lowering inventory and transport costs.

Therefore, this method shows promising results for reducing inventory and transport costs, particularly under variable demand and misalignment between loading configurations and container types.

5.8 Conclusions

We herein propose a MILP model, 3D-SCPTOP, to tackle procurement transportation planning at an operational level by incorporating 3D loading constraints into its considerations. Given the NP-hard nature of 3D bin packing, we have also developed a metaheuristic approach, Meta-3D-SCPTOP, which is later refined and enhanced by employing a mathheuristic technique, MatH-3D-SCPTOP.

A case study has also been conducted within an automotive SC that comprises a car assembler, a first-tier supplier and a second-tier supplier. The objective of the 3D-SCPTOP model is twofold: (i) to minimise the total number of trucks used; (ii) to minimise inventory holding costs. This approach takes into account the complexities of supply chain logistics in the automotive industry by demonstrating a successful integration of procurement transport planning and 3D loading constraints.

Managerial implications in a SC context are oriented to the proposed solution methodology, improving the solutions found by the company's procedure. It is also able to adapt to demand changes while finding high-quality solutions in reasonable computational times.

Nevertheless, several challenges remain for further research lines to deal with; for instance, the inclusion of multiple suppliers to better represent complex supply chains. A more comprehensive transport operation model could be achieved by incorporating routing aspects, such as vehicle routing with 3D loading constraints. Furthermore, the introduction of dynamic stability into loading constraints may also yield more practical, robust solutions. Integrating sourcing and delivery transport decisions could improve supply chain efficiency. Finally, the incorporation of production scheduling into the model has the potential of offering a cohesive approach to supply chain planning optimisation by aligning production timelines with transportation and inventory strategies.

Chapter 6

Service Network Design for RTI Transport Optimization

This paper proposes a new mixed integer linear mathematical programming (MILP) model, dubbed as SND-RTI (Service Network Design for Returnable Transport Items), for procurement transport planning in manufacturing supply chains. The model integrates transport decisions while accounting for managing returnable transport items (RTIs). Thus, we optimise the strategic decisions in terms of allocated RTIs, purchased RTIs, direct and return routes and Q-days. The optimisation model aims to minimise transportation and inventory costs. Regarding inventory and transport efficiency, the results of the 3D-SCPTOP model are superior to the company procedures.

Keywords— Procurement, inventory, transport planning, full-truckload, 3D container loading

6.1 Introduction

6.2 Literature Review

6.2.1 Research Gaps & Methodology

6.3 Problem Statement

Managing returnable transport items (RTIs) in a manufacturing supply chain is, at its core, a closed-loop logistics problem. Raw materials, parts and finished goods flow in one direction, from suppliers to customers, but the RTIs that carry them must flow back. The research question this paper addresses is the effectiveness of a collaborative return strategy of RTIs in a given SC, in which empty RTIs are rerouted and consolidated along the network, analyzing economic and environmental costs. To answer this, we propose an optimization model that determines the optimal return routes for empty RTIs, the shipment frequency on each route,

and the minimum RTI pool to prevent production disruptions, thereby defining the service network design for RTIs.

We model the SC as a directed graph $G = (V, E)$, where each node $i \in V$ represents a manufacturing site and each arc $(i, j) \in E$ represents a potential transport route, both for empty and full RTIs. Figure 6.1 illustrates a SC network, where nodes are manufacturing sites and the black arcs represent the daily average flows of full RTIs driven by production demand. Each product family requires a specific RTI type, and the number of RTIs needed per shipment depends on how many raw materials, parts or finished goods fit in a single RTI, as shown in Figure 6.2.

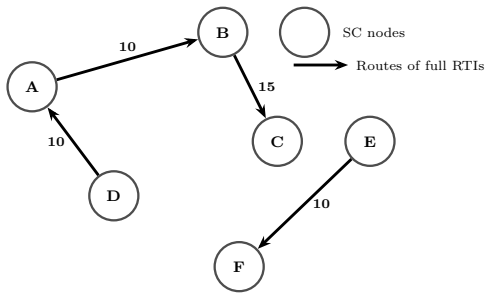


Figure 6.1: Full RTIs routes

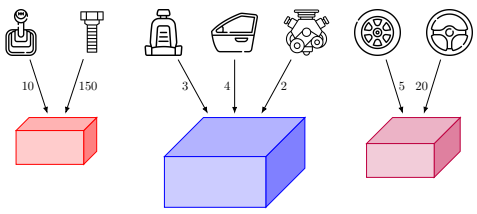


Figure 6.2: Product quantities per RTI

An RTI exists in one of two states: *full*, when loaded with raw materials, parts and finished goods, or *empty*, once its contents have been consumed at the destination site. Some RTIs can be folded, reducing their volume to decrease transport costs, as shown in Figure 6.3. The simplest RTI return strategy is a one-to-one relationship, i.e., for every shipment of full RTIs a supplier sends to a customer, the customer must return the empty RTIs along the same route, as depicted in Figure 6.4. This strategy is operationally straightforward, since each pair of nodes only coordinates with each other. However, it is rarely cost-optimal in large SCs, as it ignores consolidation opportunities along the SC and tends to result in both excessive empty RTI transport costs and a larger RTI pool than strictly necessary, which implies higher RTI investment and inventory costs.

When a central decision-maker, such as a dominant tier-1 manufacturer, an original equipment manufacturer (OEM) that owns the RTI pool, or a SC platform acting as manufacturing as a service (MaaS) Add reference, has visibility over the entire network, a collaborative RTI return strategy becomes possible. Rather than using a one-to-one RTI return

Figure 6.3: Full and empty folded RTIs

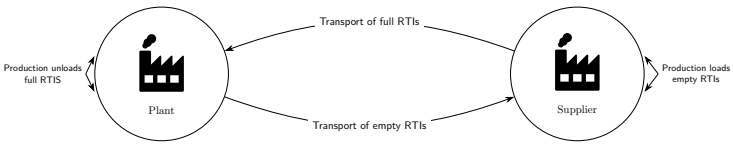
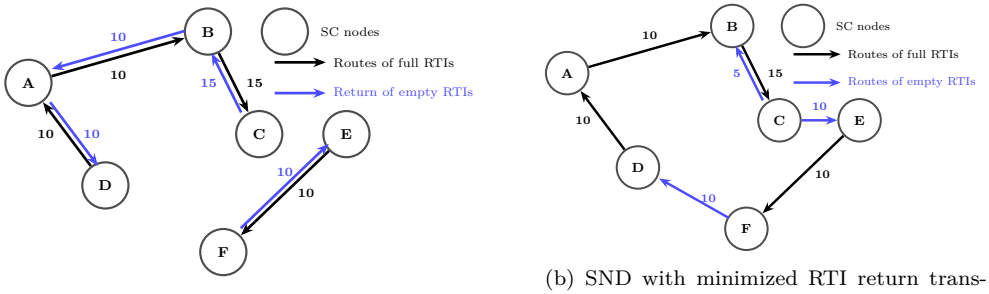


Figure 6.4: One-to-one RTI return strategy

strategy, the model optimizes empty RTI routes, reducing the total distance an empty RTI journeys and consolidating routes. This simultaneously diminishes RTI return costs and the overall RTI pool size. Figure 6.5a shows the one-to-one baseline, while Figure 6.5b illustrates how an optimized collaborative strategy reduces RTI return transport costs on the same network. Determining the service network design is the primary objective of this research.



(a) *One-to-one* relationship

(b) SND with minimized RTI return transport cost

Figure 6.5: SND with minimized RTI return transport cost

On certain routes, full RTIs are accompanied by protective accessories, such as nets or lids, that secure parts during transit. These *inlays* are themselves reusable and must be returned to the origin site after unloading. Here the FTL and LTL, also known as groupage, transport modes are addressed.

The model takes as input data the average daily flows of full RTIs between nodes, derived from released production plans over a planning horizon chosen by the decision-maker,

typically a month, quarter, or year. These averages are assumed to be representative of the considered planning horizon, meaning that demands are sufficiently stable for the average to serve as reliable input data, without extreme imbalances that would render it misleading.

While the physical routing of full RTIs is set in advance by established production and procurement contracts (i.e., a specific supplier must deliver to a specific customer a given set of products), the logistical execution of these forward flows is subject to optimization. Specifically, the model determines two operational parameters for the forward flows. First, it selects the optimal RTI type from a set of compatible options for each product. This selection affects the required transport capacity and the subsequent volume of empty returns, as different RTI types possess distinct dimensions and folding ratios. Second, the model determines the dispatch frequency for these full shipments. The shipment frequency governs the accumulation of inventory, creating a fundamental trade-off: longer intervals between shipments improve vehicle utilization and reduce fixed transport costs, but need a larger RTI pool to cover in-transit and waiting stock. Consequently, minimizing the total system cost requires the joint optimization of forward dispatch frequencies and packaging types, alongside the consolidation of empty return routes.

Under this assumption, the problem can be considered a service network design for RTIs, which consists of finding the tradeoff among daily RTI transport cost, which include transport costs for both full and empty RTIs; the inventory holding cost of the RTI pool; and CO_2 emissions from RTI transport. The model determines:

1. Routes for empty RTI and inlay return flows.
2. The RTI type selection for products with multiple compatible packaging options.
3. The shipment frequency per route (both full and empty flows), expressed as a *queue time* (Q-day), i.e. the number of days between consecutive dispatches.
4. The transport mode per route, i.e. FTL or LTL.
5. The total number of RTIs needed in the whole SC network.

The problem encompasses a strategic-tactical decision level, using as input data the daily averages of RTI demand flows. In practice, the daily average flow of full RTIs between nodes is managed by medium-to-long-term production contracts and procurement agreements, which should remain stable over the planning horizon under consideration. The Q-days and transport modes are likewise set with a medium-to-long-term frequency rather than renegotiated daily or weekly. By optimizing strategic-tactical decisions for a given planning horizon, instead of multi-period decisions, the model is computationally efficient even for large industrial networks, where a fully operational multi-period optimization could be intractable to solve. Here, the strategic-tactical perspective provides a scalable upper bound on the achievable savings from collaborative RTI management, by establishing transport routes, Q-days and the RTI pool size, which can subsequently serve as input data for operational-level RTI transport planning.

6.4 Model Formulation

Table 6.1 defines the sets, parameters and variables of the model.

Table 6.1: Model nomenclature

Indices and sets	
V	Set of nodes representing manufacturing plants; $i, j \in V$
$E \subseteq V \times V$	Set of directed edges representing potential transport routes
R	Set of RTI types; $r \in R$
$\mathcal{M}$	Set of transport modes; $m \in \mathcal{M}$
$H \subseteq V$	Set of hub nodes
Parameters	
ric_{ijpr}	Fraction of full shipments on route (i, j) of product p in RTI type r that require a return inlay; independent of transport mode
d_{ijp}	Daily demand of product p on route (i, j), in physical units (parts/day)
κ_{pr}	Capacity: number of units of product p that fit in one RTI of type r
bp_{ijp}	1 if product p must be transported on route (i, j); 0 otherwise
bf_{pr}	1 if product p is compatible with RTI type r; 0 otherwise
ric_{pr}	Fraction of full shipments of product p in RTI type r that require a return inlay
v_r^f	Volume occupied by one full (or inlay) RTI of type r
v_r^e	Volume occupied by one empty (folded) RTI of type r
be_{ijr}	1 if empty RTIs of type r may traverse route (i, j) by mode m; 0 otherwise
bm_{ijm}	1 if mode m is available on route (i, j); 0 otherwise
$c_{ijm}^{\$}$, $c_{ijm}^{CO_2}$	Variable transport cost and emissions per unit volume on route (i, j) by mode m
$f_{ijm}^{\$}$, $f_{ijm}^{CO_2}$	Fixed dispatch cost and emissions per shipment on route (i, j) by mode m
v^f, v^e	Volume occupied by one full (or inlay) RTI and one empty RTI, respectively
l_{ijm}	Transit time on route (i, j) by mode m, in days
$q_m^{\min}$, $q_m^{\max}$	Minimum and maximum queue time for mode m, in days
α^f, α^e	Maintenance loss factor applied to full/inlay and empty RTIs, respectively
ssf_{ijpr}^i, ssf_{ijpr}^j	Safety stock of full RTIs of type r for product p held at the origin and destination of route (i, j), respectively; independent of transport mode
sse_{ijpr}^i, sse_{ijpr}^j	Safety stock of empty RTIs of type r for product p held at the origin and destination of route (i, j), respectively; independent of transport mode
ω_{ijm}, δ_{ijm}	Minimum and maximum shipment volume on route (i, j) by mode m
$fhc_i^{\$}$, $fhc_i^{CO_2}$	Fixed activation cost and emissions at hub $i \in H$

$vhc_i^{\$}$, $vhc_i^{CO_2}$	Variable handling cost and emissions per unit volume at hub $i \in H$
s_r^{RTI}	Existing pool of RTI type r available before optimisation
$p_r^{\text{RTI},\$}$, p_r^{RTI,CO_2}	Unit purchase cost and embodied emissions of purchasing one RTI of type r
M	Sufficiently large constant (big-M)
Decision variables — binary	
X_{ijmr}	1 if route (i,j) carries any RTIs of type r for product p by mode m; 0 otherwise
XF_{ijmpr}, XI_{ijmr}, XE_{ijmr}	1 if route (i,j) carries full RTIs, return inlays, or empty RTIs (resp.) of type r for product p by mode m; 0 otherwise
XH_i	1 if hub $i \in H$ is activated; 0 otherwise
Decision variables — daily flow	
N_{ijmr}, NF_{ijmpr}, NI_{ijmr}, NE_{ijmr}	Daily average number of RTIs (total), full RTIs, return inlays, and empty RTIs on route (i,j), respectively
V_{ijmr}, VF_{ijmpr}, VI_{ijmr}, VE_{ijmr}	Daily average volume of RTIs (total), full RTIs, return inlays, and empty RTIs on route (i,j), respectively
Decision variables — per shipment	
q_{ijm}	Queue time on route (i,j) by mode m: days between consecutive dispatches
NS_{ijmr}, NSF_{ijmr}, NSI_{ijmr}, NSE_{ijmr}	Total number of RTIs (total), full RTIs, return inlays, and empty RTIs loaded per shipment on route (i,j)
VS_{ijmr}, VSF_{ijmr}, VSI_{ijmr}, VSE_{ijmr}	Total volume (total), full RTI volume, inlay volume, and empty RTI volume per shipment on route (i,j)
Decision variables — RTI pool	
TNC_{ijmr}, $TNCF_{ijmr}$, $TNCE_{ijmr}$, $TNCI_{ijmr}$	RTIs (total), full RTIs, empty RTIs, and inlay RTIs allocated to route (i,j), covering queue stock, in-transit stock, safety stock, and losses
P_r	Additional RTIs of type r to purchase; $P_r \geq 0$
Decision variables — cost and hub	
$TCS_{ijm}^{CO_2}$, $TCS_{ijm}^{\$}$	Transport cost and emissions per shipment on route (i,j) by mode m
$TC_{ijm}^{CO_2}$, $TC_{ijm}^{\$}$	Daily transport cost and emissions on route (i,j) by mode m
$CH_i^{CO_2}$, $CH_i^{\$}$	Daily handling cost and emissions at hub $i \in H$
$PC_r^{CO_2}$, $PC_r^{\$}$	Amortized daily purchase cost and emissions for RTI type $r \in R$

Variables in the model carry up to five indices (see Table 6.1). To avoid writing three and four-level summations and writing additional constraints to encode Boolean logic, we adopt the following convention: suppressing one or more indices from a variable and placing an accent on its name produces a derived variable whose values are obtained by collapsing the suppressed indices according to one of three aggregation rules: summation ($\bar{\ }$), logical OR ($\check{\ }$), or at-most-one ($\hat{\ }$). The accent identifies which rule is applied.

Convention 1 (Index suppression and aggregation).

We call the original, fully-indexed variable the *root variable*. Each index of a root variable ranges over a set. The sets over which the retained indices range are called the *projected sets*, and their Cartesian product is denoted Π; the sets over which the suppressed indices

range are called the *suppressed sets*, and their Cartesian product is denoted $\mathcal{S}$. We write $\pi \in \Pi$ for a generic element of the projected domain and $\sigma \in \mathcal{S}$ for a generic element of the suppressed domain. A derived variable is defined for every $\pi \in \Pi$ by collapsing all $\sigma \in \mathcal{S}$ according to the aggregation rule indicated by the accent.

Whenever a derived variable appears in any constraint of the model, the variable and its defining constraints (as specified by the applicable rule below) are automatically introduced into the formulation for every $\pi \in \Pi$. In other words, writing a derived variable in a constraint is a shorthand that simultaneously declares the variable and appends its linking constraints to the model; no separate definition is needed.

Rule 1 (Sum, $\bar{\ }$). Suppressing one or more indices and writing a $\bar{\ }$ over the root variable denotes the sum over the suppressed domain $\mathcal{S}$. For every $\pi \in \Pi$ the derived variable is defined by:

$$\overline{\phi}_\pi \;:=\; \sum_{\sigma \in \mathcal{S}} \phi_{\pi,\sigma} \quad \forall\, \pi \in \Pi.$$

For continuous and integer root variables the result is standard summation; for binary root variables it represents an integer count of active entries, not a Boolean sum.

Example. From the root variable NF_{ijmpr} $\big((i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ \ r \in R\big)$, suppressing p ($\Pi = E \times \mathcal{M} \times R$, $\mathcal{S} = P$) and suppressing all indices ($\Pi = \varnothing$, $\mathcal{S} = E \times \mathcal{M} \times P \times R$):

$$\overline{NF}_{ijmr} \;=\; \sum_{p \in P} NF_{ijmpr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ r \in R, \tag{6.1}$$

$$\overline{NF} \;=\; \sum_{(i,j) \in E} \sum_{m \in \mathcal{M}} \sum_{p \in P} \sum_{r \in R} NF_{ijmpr}. \tag{6.2}$$

Rule 2: Positional disambiguation. When two indices range over the same set, as $i, j \in V$ do in NF_{ijmpr}, suppressing one by name alone is ambiguous. A dot $(\cdot)$ marks the suppressed position: $\overline{NF}_{\cdot jmpr} := \sum_{i \in V} NF_{ijmpr}$ suppresses i, while $\overline{NF}_{i \cdot mpr} := \sum_{j \in V} NF_{ijmpr}$ suppresses j. This convention applies to all three rules below.

Rule 3 (OR, $\breve{\ }$). **Binary root variables only.** Suppressing one or more indices and writing a $\breve{\ }$ over the root variable symbol introduces a new binary variable equal to 1 if and only if at least one root binary sharing the same projected element π is active. For every $\pi \in \Pi$ the new variable $\breve{\phi}_\pi$ is coupled to the corresponding sum $\overline{\phi}_\pi$ (Rule 1) via:

$$\breve{\phi}_\pi \le \overline{\phi}_\pi \qquad \forall\, \pi \in \Pi, \tag{6.3}$$

$$\overline{\phi}_\pi \le |\mathcal{S}|\ \breve{\phi}_\pi \qquad \forall\, \pi \in \Pi, \tag{6.4}$$

where $|\mathcal{S}|$ is the cardinality of the suppressed domain. Together, (6.3)–(6.4) enforce $\breve{\phi}_\pi = 1 \iff \overline{\phi}_\pi \ge 1$ and, equivalently, $\breve{\phi}_\pi = 0 \iff \overline{\phi}_\pi = 0$.

Example. From the binary XF_{ijmpr}, suppressing the indices m, p, and r we get ($\Pi = E$, $\mathcal{S} = P \times \mathcal{M} \times R$, $|\mathcal{S}| = |R||\mathcal{M}||P|$) yields $\breve{XF}_{ij}$, which equals 1 whenever route (i,j) carries full RTIs:

$$\breve{XF}_{ij} \le \overline{XF}_{ij} \qquad \forall\, (i,j) \in E, \tag{6.5}$$

$$\overline{XF}_{ij} \le |R||\mathcal{M}||P|\ \breve{XF}_{ij} \qquad \forall\, (i,j) \in E. \tag{6.6}$$

Rule 4 (At-most-one, $\widehat{\ }$). **Binary root variables only.** Suppressing one or more indices and writing a $\widehat{\ }$ over the root variable symbol introduces a new binary variable that equals the sum and simultaneously caps it at 1, asserting that at most one root binary sharing the same projected element π may be active. For every $\pi \in \Pi$ the derived variable is coupled to the corresponding sum via:

$$\widehat{\phi}_\pi \leq \overline{\phi}_\pi \qquad \forall\, \pi \in \Pi, \tag{6.7}$$

$$\overline{\phi}_\pi \leq \widehat{\phi}_\pi \qquad \forall\, \pi \in \Pi. \tag{6.8}$$

Because $\widehat{\phi}_\pi \in \{0,1\}$, both inequalities together force $\overline{\phi}_\pi \in \{0,1\}$: either no root binary is active ($\widehat{\phi}_\pi = 0$) or exactly one is ($\widehat{\phi}_\pi = 1$). This is equivalent to an SOS1 constraint on the group of root binaries indexed by $\mathcal{S}$ that share the same projected element π.

Example. From XF_{ijmpr}, suppressing m and r ($\Pi = E \times P$, $\mathcal{S} = \mathcal{M} \times R$) yields $\widehat{XF}_{ijp}$, which enforces that at most one mode–RTI-type pair may carry product p on route (i,j):

$$\widehat{XF}_{ijp} \leq \overline{XF}_{ijp} \qquad \forall\, (i,j) \in E,\ p \in P, \tag{6.9}$$

$$\overline{XF}_{ijp} \leq \widehat{XF}_{ijp} \qquad \forall\, (i,j) \in E,\ p \in P. \tag{6.10}$$

Table 6.2 lists the derived variables used in the model. The accent identifies the aggregation rule; the suppressed-indices column shows which dimensions are collapsed. Each variable and its linking constraints are introduced automatically at the point of first use.

Table 6.2: Derived variables used in the model.

Derived	Root	Suppr.	Rule	Interpretation
$\overline{NF}_{ijmr}$	NF_{ijmpr}	p	$\overline{\ }$	Total daily full-RTI flow on (i,j) by mode m, type r
$\overline{N}_{i\cdot r}$	N_{ijmr}	j,m	$\overline{\ }$	Daily outflow of RTI type r from node i †
$\overline{N}_{\cdot ir}$	N_{ijmr}	j,m	$\overline{\ }$	Daily inflow of RTI type r into node i †
$\widetilde{XF}_{ijmr}$	XF_{ijmpr}	p	$\widetilde{\ }$	Any product uses full RTIs on (i,j) by m,r?
$\check{X}_{ijm}$	X_{ijmr}	r	$\widetilde{\ }$	Mode m active on (i,j) for any RTI type?
$\check{X}_{i\cdot}$	X_{ijmr}	j,m,r	$\widetilde{\ }$	Node i dispatches any flow? (hub activation)
$\widetilde{XE}_{i\cdot r}$, $\widetilde{XE}_{\cdot ir}$	XE_{ijmr}	j,m	$\widetilde{\ }$	Node i dispatching / receiving empties of type r?
$\widehat{XF}_{ijp}$	XF_{ijmpr}	m,r	$\widehat{\ }$	At most one mode–RTI-type pair for product p on (i,j)

†In the outflow variable $\overline{N}_{i\cdot r}$ the first index of the root N_{ijmr} is the focal node i and the suppressed dot replaces j; in the inflow variable $\overline{N}_{\cdot ir}$ the second index is the focal node i and the dot replaces the first index. Both derive from the same root N_{ijmr}, with the positional role of i swapped.

Objective functions. The model evaluates two conflicting objectives: the total economic cost ($Z^{\$}$) and the total environmental impact in terms of CO_2 emissions (Z^{CO_2}). Both objectives account for transport, RTI purchase, and hub handling operations:

$$\min \quad Z^{\$} = \overline{TC}^{\$} + \overline{PC}^{\$} + \overline{CH}^{\$} \tag{6.11}$$

$$\min \quad Z^{CO_2} = \overline{TC}^{CO_2} + \overline{PC}^{CO_2} + \overline{CH}^{CO_2} \tag{6.12}$$

Flow conservation. RTIs circulate in a closed loop: the total average daily outflow of RTI type r from any node must equal its total average daily inflow:

$$\overline{N}_{i\cdot r} = \overline{N}_{\cdot ir} \qquad \forall\, i \in V,\ r \in R, \tag{6.13}$$

Flow–binary consistency. Each root binary is active if and only if the corresponding daily average flow is strictly positive, and only when the route is physically permissible as encoded by the b-parameters:

$$NF_{ijmpr} \leq M \cdot XF_{ijmpr}, \quad XF_{ijmpr} \leq NF_{ijmpr} \cdot bf_{ijp} \cdot bm_{ijm} \cdot bp_{pr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R \tag{6.14}$$

$$NI_{ijmpr} \leq M \cdot XI_{ijmpr}, \quad XI_{ijmpr} \leq NI_{ijmpr} \cdot bf_{jip} \cdot bm_{ijm} \cdot bp_{pr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R \tag{6.15}$$

$$NE_{ijmr} \leq M \cdot XE_{ijmr}, \quad XE_{ijmr} \leq NE_{ijmr} \cdot be_{ijr} \cdot bm_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R. \tag{6.16}$$

$$N_{ijmr} = \overline{NF}_{ijmr} + \overline{NI}_{ijpr} + NE_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R. \tag{6.17}$$

Route activation. A route is active if and only if it carries at least one RTI in any state: full, empty, or inlay:

$$3 \cdot X_{ijmr} \geq \breve{XF}_{ijmr} + XE_{ijmr} + \breve{XI}_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.18}$$

$$X_{ijmr} \leq \breve{XF}_{ijmr} + XE_{ijmr} + \breve{XI}_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R. \tag{6.19}$$

Full-route activation. For any given product p that has to be transported from i to j, only one RTI type r can be used. $\widehat{XF}_{ijp}$ therefore equals 1 if and only if product p is used in route (i,j), as encoded by the binary parameter bp_{ijp}:

$$\widehat{XF}_{ijp} = bf_{ijp} \qquad \forall\,(i,j) \in E,\ p \in P. \tag{6.20}$$

This simultaneously enforces demand-driven activation and the uniqueness of RTI type per product per route.

Transport mode selection. A mode may only be used on a route if it is physically available there, and at most one mode may operate on any given route:

$$\check{X}_{ijm} \leq bm_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M} \tag{6.21}$$

$$\sum_{m \in \mathcal{M}} \check{X}_{ijm} \leq 1 \qquad \forall\,(i,j) \in E, \tag{6.22}$$

Hub nodes. Hub nodes $i \in H \subseteq V$ are intermediate consolidation points that aggregate outgoing flows and incur a handling cost with fixed component $fhc_i \geq 0$ and variable component $vhc_i \geq 0$ per unit volume.

At every non-hub node $i \in V \backslash H$, a node may not simultaneously receive and dispatch empty RTIs of the same type $r \in R$:

$$\breve{XE}_{\cdot ir} + \breve{XE}_{i \cdot r} \leq 1 \qquad \forall\, i \in V \backslash H, r \in R. \tag{6.23}$$

For every hub node $i \in H$, the binary XH_i equals 1 if and only if hub i is activated, i.e. if at least one outgoing route from i is used:

$$XH_i = \check{X}_{i\cdot} \qquad \forall, i \in H \tag{6.24}$$

$$CH_i^{\$} = fhc_i^{\$} \cdot XH_i + vhc_i^{\$} \cdot \overline{V}_{i\cdot} \qquad \forall\, i \in H \tag{6.25}$$

$$CH_i^{CO_2} = fhc_i^{CO_2} \cdot XH_i + vhc_i^{CO_2} \cdot \overline{V}_{i\cdot} \qquad \forall\, i \in H. \tag{6.26}$$

Full-RTI route flow. The number of full containers for a route depends on the chosen RTI type $r \in R$ and the demand of the product p:

$$NF_{ijmpr} \cdot \kappa_{pr} \geq d_{ijp} \cdot XF_{ijmpr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R. \tag{6.27}$$

Inlay return flows. Where a full shipment on route (i,j) requires an inlay, that inlay is returned on the reverse route (j,i). The daily return-inlay flow is proportional to the fraction of full shipments that require one and the RTI is the same:

$$\overline{XI}_{jipr} = \overline{XF}_{ijpr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R, \tag{6.28}$$

$$\overline{NI}_{jipr} \geq ric_{pr} \cdot \overline{NF}_{ijpr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R. \tag{6.29}$$

Daily average volumes. Full and inlay RTIs are carrying items and occupy volume v^f per unit; empty RTIs are collapsed and occupy volume v^e per unit. The total daily volume on each route is the sum of the three:

$$VF_{ijmpr} = NF_{ijmpr} \cdot v^f \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.30}$$

$$VI_{ijmpr} = NI_{ijmpr} \cdot v^f \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.31}$$

$$VE_{ijmr} = NE_{ijmr} \cdot v^e \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.32}$$

$$V_{ijmr} = \overline{VF}_{ijmr} + \overline{VI}_{ijmr} + VE_{ijmr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R. \tag{6.33}$$

Per-shipment RTI counts. Shipments are made with a regular frequency denoted by q_{ijm}. The number of RTIs loaded onto each shipment equals the daily average flow multiplied by the queue time. These products are *nonlinear* and will be linearised in Section **??**:

$$NSF_{ijmpr} = NF_{ijmpr} \cdot q_{ijm} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.34}$$

$$NSI_{ijmpr} = NI_{ijmpr} \cdot q_{ijm} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.35}$$

$$NSE_{ijmr} = NE_{ijmr} \cdot q_{ijm} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.36}$$

$$NS_{ijmr} = NSF_{ijmr} + NSI_{ijmr} + NSE_{ijmr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R. \tag{6.37}$$

Per-shipment volumes. Analogously, per-shipment volumes are the daily average volumes scaled by the queue time, and are likewise nonlinear:

$$VSF_{ijmpr} = VF_{ijmpr} \cdot q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.38}$$

$$VSI_{ijmpr} = VI_{ijmpr} \cdot q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.39}$$

$$VSE_{ijmr} = VE_{ijmr} \cdot q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.40}$$

$$VS_{ijmr} = \overline{VSF}_{ijmr} + \overline{VSI}_{ijmr} + VSE_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R. \tag{6.41}$$

Queue time bounds. The queue time must lie within the operationally permitted range $[q_m^{\min}, q_m^{\max}]$ for the chosen mode; on inactive routes the queue time collapses to zero:

$$q_m^{\min} \cdot \check{X}_{ijm} \;\le\; q_{ijm} \;\le\; q_m^{\max} \cdot \check{X}_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M}. \tag{6.42}$$

Shipment volume bounds. Each active shipment must meet a minimum and maximum load factor of the selected mode:

$$\omega_{ijm} \cdot \check{X}_{ijm} \;\le\; \overline{VS}_{ijm} \;\le\; \delta_{ijm} \cdot \check{X}_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M}, \tag{6.43}$$

Total RTI pool requirements. The RTIs allocated to a route cover four components: the queue stock loaded per shipment; the in-transit stock en route; the safety stocks held at both endpoints; and a margin for maintenance losses. Safety stocks depend on the route, product, and RTI type only, and do not vary by mode. Full and inlay RTIs attract a loss factor of α^f; empty RTIs attract α^e. We do not contemplate the necessesity of considering safety stocks for inlay RTIs.

$$TNCF_{ijmr} = (1+\alpha^f)\Big(NSF_{ijmr} + l_{ijm} \cdot NF_{ijmr} + XF_{ijmr}\big(ssf^i_{ijpr} + ssf^j_{ijpr}\big)\Big) \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.44}$$

$$TNCE_{ijmr} = (1+\alpha^e)\Big(NSE_{ijmr} + l_{ijm} \cdot NE_{ijmr} + XE_{ijmr}\big(sse^i_{ijpr} + sse^j_{ijpr}\big)\Big) \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.45}$$

$$TNCI_{ijmr} = (1+\alpha^f)(NSI_{ijmr} + l_{ijm} \cdot NI_{ijmr}) \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R \tag{6.46}$$

$$TNC_{ijmr} = TNCF_{ijmr} + TNCE_{ijmr} + TNCI_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p,\ r \in R. \tag{6.47}$$

RTI purchase. If the total pool requirement across all routes exceeds the existing stock of RTI type r, the shortfall must be procured:

$$\overline{TNC}_r \;\le\; s_r^{\mathrm{RTI}} + P_r, \qquad P_r \ge 0 \qquad \forall\, r \in R. \tag{6.48}$$

Daily transport costs and emissions. The cost and emissions of a single shipment on route (i,j) by mode m combine a variable charge proportional to total shipment volume and a fixed dispatch value incurred whenever the route is active. Converting to a daily equivalent by dividing by the queue time yields the daily route impact; this division is *nonlinear* and

will be linearised in Section **??**:

$$TCS^{\$}_{ijm} = VS_{ijm} \cdot c^{\$}_{ijm} + \check{X}_{ijm} \cdot f^{\$}_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M} \tag{6.49}$$

$$TCS^{CO_2}_{ijm} = VS_{ijm} \cdot c^{CO_2}_{ijm} + \check{X}_{ijm} \cdot f^{CO_2}_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M} \tag{6.50}$$

$$TC^{\$}_{ijm} \cdot q_{ijm} = TCS^{\$}_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M} \tag{6.51}$$

$$TC^{CO_2}_{ijm} \cdot q_{ijm} = TCS^{CO_2}_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M}. \tag{6.52}$$

6.5 Computational Experiments

This section evaluates the proposed SND-RTI model through a series of computational experiments on randomly generated instances that approximate real-world manufacturing supply chains. The experiments pursue two objectives: (i) assessing the computational scalability of the linearised MILP as the network grows; and (ii) quantifying the cost savings of the optimised collaborative return strategy relative to the conventional one-to-one baseline.

6.5.1 Instance Generation

Instances are generated to mirror the structure of multi-tier manufacturing supply chains, where plants are geographically clustered in industrial zones and connected by transport routes of varying distance and mode availability.

Network topology. Each instance is built in three steps.

1. **Industrial zones.** n_Z industrial zones are placed uniformly at random in a $[0, L] \times [0, L]$ km^2 area, where L is chosen according to the instance class (Table 6.3). Each zone z contains n^L_z large plants and n^S_z small plants, drawn from the ranges specified in Table 6.3. Plants within a zone are scattered around the zone centre with a Gaussian offset (σ_{intra} km).
2. **Route generation.** A full-RTI route (i, j) is generated between every ordered pair of plants that satisfies a demand condition (see below). Large plants have a higher probability p^{inter} of sourcing from plants in *other* zones, while small plants have a higher probability p^{intra} of sourcing within the *same* zone. Specifically, for each plant j, a set of suppliers is sampled: with probability p^{inter} (resp. p^{intra}) a candidate supplier is drawn from another zone (resp. the same zone). The expected number of supply routes per plant is controlled by parameter $\bar{k}$. Empty-RTI return routes are allowed between any pair of nodes in the network, i.e. $be_{ijmr} = 1$ for all $(i, j) \in V \times V$, $m \in \mathcal{M}$, $r \in R$, since the model optimises which of these potential return arcs are actually used.
3. **Hub designation.** Each industrial zone contains n_H candidate hub nodes ($H \subseteq V$), reflecting the common practice of co-locating consolidation centres within industrial areas. The value of n_H scales with the instance class: $n_H = 1$ for Small, $n_H = 2$ for Medium, and $n_H = 3$ for Large instances (see Table 6.3). Within each zone, hub candidates are selected among the nodes with the highest total inflow plus outflow of full RTIs. The total number of candidate hubs is therefore $|H| = n_H \cdot n_Z$; the model decides which of these candidates to activate.

Demand and product assignment. Each full-RTI route (i, j) carries one or more products from the product set P. The number of products per route depends on the instance class and the type of the receiving plant: routes serving large plants carry up to 2 products in Small instances, up to 3 in Medium, and up to 4 in Large instances; routes serving small plants carry a single product. For each product assigned to a route, the daily average demand in full RTI units, μ_{ijmpr}, is drawn from a log-normal distribution $\text{LogNormal}(\mu_d, \sigma_d)$, rounded to the nearest integer, to reflect the right-skewed demand patterns typical of manufacturing supply chains. Large plants receive demands from the upper quartile of the distribution to model their higher throughput. The inlay fraction ric_{ijpr} is drawn uniformly from $[0, r^{\max}]$.

RTI types and transport modes. Each product p is compatible with a subset of RTI types $R_p \subseteq R$. The majority of products are assigned to a single RTI type ($|R_p| = 1$), chosen uniformly at random from R. A smaller fraction of products admits multiple compatible types: up to 2 in Small instances, 2–3 in Medium, and up to 4 in Large instances. The parameter bf_{ijmrp} is set to 1 for every compatible product–RTI-type pair on routes where product p is demanded; the model selects the type to use. Two transport modes are considered throughout: $\mathcal{M} = \{\text{FTL}, \text{LTL}\}$. Mode availability bm_{ijm} depends on distance: routes shorter than $d^{\text{FTL}}_{\min}$ km are LTL-only; routes longer than $d^{\text{LTL}}_{\max}$ km are FTL-only; all others admit both modes.

Cost and capacity parameters. Variable transport costs c_{ijm} are set proportional to the Euclidean distance between nodes, with mode-specific rates c^{FTL} and c^{LTL} per km per m^3. Fixed dispatch costs f_{ijm} are mode-dependent constants. Vehicle capacities δ_{ijm} and minimum load factors ω_{ijm} follow industry-standard values for full-truckload and groupage shipments. Transit times l_{ijm} are obtained by dividing the Euclidean distance by an average speed of $v_{\text{avg}} = 500$ km/day, common to both modes. Mode availability thresholds are set to $d^{\text{FTL}}_{\min} = 50$ km and $d^{\text{LTL}}_{\max} = 800$ km.

Safety stocks and existing pool. Safety stocks at both endpoints of each route are set to three days of the daily average demand on that route: $ssf^i_{ijpr} = ssf^j_{ijpr} = 3\,\mu_{ijmpr}$ for full RTIs, and $sse^i_{ijpr} = sse^j_{ijpr} = 3\,NE_{ijmr}$ for empty RTIs. The existing RTI pool is set to zero for all types ($s^{\text{RTI}}_r = 0$), so that the model must purchase the entire pool; this isolates the effect of the network design on total RTI investment. Non-zero initial pools are reserved for the case study, where historical inventory data are available.

Linearisation. The continuous queue time q_{ijm} is discretised over the set of admissible integer values $\mathcal{Q}_m = \{q^{\min}_m, q^{\min}_m+1, \ldots, q^{\max}_m\}$, yielding $|\mathcal{Q}_{\text{FTL}}| = 8$ and $|\mathcal{Q}_{\text{LTL}}| = 5$ candidate values. Each candidate $k \in \mathcal{Q}_m$ is associated with a binary variable indicating whether it is selected; all bilinear products of the form $q_{ijm} \cdot (\cdot)$ are replaced by their linearised counterparts as described in the model formulation.

RTI volumes v^f_r and v^e_r, purchase costs p^{RTI}_r, loss factors α^f and α^e, and hub costs fhc_i and vhc_i are sampled from the ranges reported in Table 6.4.

6.5.2 Instance Classes

Three classes of instances—Small, Medium, and Large—are defined by scaling the number of industrial zones, plants per zone, RTI types, and products. Table 6.3 summarises the dimensions. For each class, N_{inst} instances are generated with different random seeds to allow statistical comparison.

Table 6.3: Instance classes and their dimensions.

Class	n_Z	n_z^L	n_z^S	n_H	$\lvert V\rvert$	$\lvert E\rvert$ (approx.)	$\lvert R\rvert$	$\max \lvert R_p\rvert$	$\lvert P\rvert$	$\lvert\mathcal{M}\rvert$	N_{inst}
Small	2–3	1–2	2–4	1	10–20	15–40	2	2	3–5	2	10
Medium	4–6	2–3	4–8	2	30–60	60–150	3–4	3	6–10	2	10
Large	8–12	3–5	6–12	3	80–180	200–600	4–6	4	10–20	2	10

6.5.3 Experimental Setup

All experiments are run on a workstation equipped with an Intel® Core™ i7-XXXXX processor (X cores, X.X GHz) and XX GB RAM, running Ubuntu XX.XX. The linearised MILP is solved with Gurobi XX.X using default settings, with a time limit of 3 600 s for Small and Medium instances and 7 200 s for Large instances. The optimality gap tolerance is set to 0.5 %.

6.5.4 Experiment 1: Computational Scalability

The first experiment evaluates how the solution time and the optimality gap scale with the instance dimensions. For each instance class, Table 6.5 reports the number of binary and continuous variables, the number of constraints, the average solution time, the median optimality gap at termination, and the fraction of instances solved to the gap tolerance within the time limit.

To identify the dominant source of complexity, we additionally report solve times as a function of each individual scaling dimension—$|V|$, $|E|$, and $|R|$—while holding the other two approximately constant.

6.5.5 Experiment 2: Cost Savings Relative to the One-to-One Baseline

The second experiment compares the optimised SND-RTI solution with the one-to-one baseline described in Section **??**. In the baseline, every full-RTI route (i, j) is paired with a mirror return route (j, i) that ships back the exact number of empty RTIs consumed at j, using the same transport mode. No consolidation or rerouting of empty RTIs is allowed, and no hubs are activated.

For each instance, we compute the baseline cost by fixing the empty return routes to their one-to-one configuration and solving the remaining model (queue times and RTI pool) to optimality. The relative savings of the SND-RTI solution are decomposed into three components:

1. **Transport cost savings** (ΔTC): reduction in the total daily transport cost $\overline{TC}$, driven by route consolidation and mode switching on the return legs.
2. **RTI pool savings** (ΔP): reduction in the number of purchased RTIs, achieved by shorter return cycles and better pool utilisation.

3. **Hub cost** (ΔCH): the additional hub-handling cost incurred by the optimised solution, partially offsetting the above savings.

Table 6.6 reports the average and range of these savings across all instances within each class.

We further analyse how the network density—measured as the ratio $|E|/|V|$—affects the magnitude of savings, since denser networks offer more consolidation opportunities for empty-RTI returns.

6.5.6 Discussion

Table 6.4: Parameter ranges for instance generation.

Parameter	**Range / Value**	**Unit**
Network generation		
L (area side)	500 / 1 000 / 1 500	km
σ_{intra}	5–15	km
$\bar{k}$ (avg. suppliers per plant)	2–4	—
p^{inter} (large plants)	0.7	—
p^{intra} (small plants)	0.7	—
n_H (candidate hubs per zone)	1 / 2 / 3	—
Demand and products		
μ_d, σ_d (demand log-normal)	$(2.0, 0.8)$	RTI/day
$r^{\max}$ (max inlay fraction)	0.3	—
$\lvert R_p \rvert$ (compatible RTI types per product)	1 (majority); up to 2 / 3 / 4	—
Transport		
c^{FTL}, c^{LTL}	0.05, 0.12	€/km/m^3
f^{FTL}, f^{LTL}	150, 40	€/shipment
δ^{FTL}, δ^{LTL}	80, 30	m^3
ω^{FTL}, ω^{LTL}	40, 5	m^3
v_{avg} (average speed)	500	km/day
$d^{\text{FTL}}_{\min}$ (FTL minimum distance)	50	km
$d^{\text{LTL}}_{\max}$ (LTL maximum distance)	800	km
$[q^{\min}, q^{\max}]_{\text{FTL}}$	$[3, 10]$	days
$[q^{\min}, q^{\max}]_{\text{LTL}}$	$[1, 5]$	days
$\lvert \mathcal{Q}_{\text{FTL}} \rvert$, $\lvert \mathcal{Q}_{\text{LTL}} \rvert$	8, 5	—
RTI and inventory		
v^f_r	0.3–0.8	m^3
v^e_r / v^f_r (folding ratio)	0.2–0.5	—
p^{RTI}_r	30–120	€
s^{RTI}_r (existing pool)	0	—
ss_{days} (safety stock)	3	days
α^f, α^e	0.02–0.05	—
Hubs		
fhc_i	200–500	€/day
vhc_i	0.5–2.0	€/m^3

Table 6.5: Computational scalability across instance classes. Times and gaps are averaged over N_{inst} instances per class.

Class	**Bin. vars**	**Cont. vars**	**Constrs**	**Avg. time (s)**	**Med. gap (%)**	**Solved (%)**
Small	—	—	—	—	—	—
Medium	—	—	—	—	—	—
Large	—	—	—	—	—	—

Table 6.6: Cost savings of SND-RTI over the one-to-one baseline (average ± std. dev. across instances).

Class	ΔTC (%)	ΔP (%)	ΔCH (€/day)	**Total savings (%)**
Small	—	—	—	—
Medium	—	—	—	—
Large	—	—	—	—

Chapter 7

Route Clustering Algorithms for Procurement Transport

Efficiently distributing goods and materials is a vital aspect of logistics and supply chain management. In this paper, we develop a clustering mathematical programming model to consolidate the transportation logistics of an established set of routes. Each route has a determined weight and volume of transported goods per week. In this context, we study the impact of consolidating routes to procure the manufacturing facilities by leveraging the weight and volume of the established routes. To assist during the decision process, the model has different parameters that the decision maker can adjust in order to better model the supply chain under study. Furthermore, a branch and price algorithm is developed and tested to solve the real-world instances inspired in the automotive sector with up to 900 suppliers and 50 manufacturing facilities. Thus the decision tool has been validated in a first-tier supplier of the automotive industry.

Keywords— Edge clustering Supply chain Procurement Transport Branch and Price

7.1 Introduction

Maximising economic potential and social benefits, and minimising environmental impacts are critical objectives in the efficient utilisation of productive and logistics resources in manufacturing supply chains. Procurement, as defined by Chopra and Meindl, 2015, is the process by which goods and services are acquired in a supply chain, whose main aim is to source raw materials, components, or finished products for a factory or distribution centre from suppliers. Note that while the supplier "typically" assumes the responsibility for transport planning, there are cases, such as the automotive industry, in which manufacturers actively participate in making transport decisions due to the complex interactions between numerous manufacturing facilities and suppliers Fleischmann et al., 2015. This indicates that transport decisions during procurement should be a collaborative effort made among all supply chain parties to ensure that the sourcing process is efficiently managed. Here we focus on

optimising the tactical decisions involved in procurement transport planning by underscoring the importance of strategic collaboration by aggregating the transport resources of different routes to enhance supply chain efficiency.

According to Díaz-Madroñero, Peidro, and Mula, 2014, automatic algorithm-based procurement transport planning is preferable to manual planning based on planners' judgement and experience. The latter can be myopic and generate suboptimal plans. Such insight should be amplified when considering the optimisation of procurement transport problems when there are many nodes in the supply chain, where the coordination of the different stakeholders can reduce logistics, transportation costs and the environmental impact. One possible way to coordinate these logistic processes is to cluster the routes between suppliers and manufacturing facilities so that the routes in the same cluster share transportation resources. This is of outmost importance when routes are long and constrained by weight and volume because we can mix the required goods inside trucks to minimise the number of trucks. Hence this article aims to bridge this research gap by providing a novel mixed integer linear mathematical programming (MILP) model that clusters the routes between the suppliers and manufacturing facilities to minimise the procurement transport cost along the supply chain. Additionally, as a solution approach, branch and price algorithms is developed. Branch and price combine two methods: branch-and-bound and column generation (CG). It is often applied to problems like vehicle routing, crew scheduling, and cutting stock problems. CG is a technique to solve large MILP models with a determined decomposable structure. It was introduced in Ford and Fulkerson, 1958 to address a multi-commodity network flow problem. Then, Dantzig and Wolfe, 1960 adapted it to LP problems with a decomposable structure. Furthermore, Gilmore and Gomory, 1961 demonstrated the efficiency of the CG technique to the cutting stock problem. For a detailed explanation about the inner workings of CG and branch and price techniques we refer readers to Lübbecke and Desrosiers, 2005 and Desrosiers and Lübbecke, 2011. Here, a solution approach based on a branch and price algorithm using column generation is developed to solve real-world instances with up to 900 suppliers, 50 manufacturing facilities and 1,800 routes.

The main contributions of this paper are: (i) formulating a novel MILP model for the procurement transport supply chain route clustering, dubbed as SCRC, that integrates volume and weight distribution; (ii) developing a branch and price algorithm to find good quality solutions in a reasonable amount of time; (iii) validating and testing the SCRC model with a case study at a first-tier supplier of the automotive industrial sector. Readers are also referred to the literature reviews of Mosca et al., 2019 and Díaz-Madroñero et al., 2015, which focus on the integration of inventory decisions and transport decisions.

The rest of the paper is organised as follows. Section 2 presents the literature review. Section 3 describes the addressed problem. Section 4 introduces the mathematical formulation of the SCRC. Section 5 describes the branch and price procedure. Section 6 includes the computational experiments. Section 7 explains how the SCRC can be used as a decision support tool. Section 8 concludes the research and analyses future research lines.

7.2 Literature Review

Transport processes are, extensively, explored in academic research due to their critical role in supply chains. As highlighted by Langley et al., 2020, transportation is essential for transferring raw materials and final products across different supply chain stages, including suppliers, warehouses, manufacturers, and retailers. Fleischmann et al., 2015 categorise transport processes in supply chains into two main types: the transport of materials from suppliers to manufacturing facilities, known as procurement transport; the distribution of finished goods from factories to consumers. This discussion focuses on addressing issues related to procurement transport.

Procurement transport problems can be analysed with two different dimensions: the configuration of the supply chain and the used transportation method. When considering supply chain configurations, Mosca et al., 2019 distinguish among four different scenarios: (i) from a single source to a single destination (SS); (ii) from a single source to multiple destinations (SM); (iii) from multiple sources to a single destination (MS); (iv) from multiple sources to multiple destinations (MM). The "source" reflects the number of considered supplier locations, whereas the "destination" indicates the number of manufacturing sites receiving supplies. Additionally, Díaz-Madroñero, Peidro, and Mula, 2014 highlight various transportation methods, including full-truckload (FTL), less-than-truckload (LTL) and milk run. Here the aggregation of transportation resources in the scientific literature is also a main concern.

Schöneberg et al., 2010 study the concept of delivery profiles, which determine the frequency at which deliveries are made and are set at a fixed rate. In this context, FTL transport occurs when the volume of goods exceeds the capacity of one vehicle and, therefore, direct shipment is a viable option. LTL is used when the volume of goods does not fill an entire truck. In this case, goods are shipped to a consolidation centre before being transported to their final destination. The proposed model is designed for the MS supply chain configuration and routes are consolidated by LTL. In Schöneberg et al., 2013, the proposed model considers demand uncertainty.

Díaz-Madroñero et al., 2017 introduce an MILP model to integrate production, procurement and transport planning decisions into manufacturing systems. The proposed model goes beyond traditional material requirement planning systems by considering the MS supply chain configuration in a unified optimisation framework. It takes into account the capacities of material and production resources, as well as procurement transport planning decisions. Routes can be consolidated using three modes of transport: FTL, LTL and milk run.

Chitsaz et al., 2019 address integrating production, inventory and procurement transport into a manufacturing context by defining this problem as an assembly routing problem (ARP). It introduces an MILP model for the ARP and proposes a three-phase decomposition matheuristic as a solution approach. Routes are consolidated using a milk-run scheme to procure products.

Baller et al., 2022 study the optimisation of inbound transportation of an automotive factory from first-tier suppliers. The MS study uses FTL, LTL and courier express service (CES) as transport methods. Routes are consolidated using LTL as a transport method.

Bou-Zeid et al., 2023 examine a multiperiod tactical transportation plan by focusing on the inbound logistics for an automobile manufacturer. The study explores direct, indirect, and milk-run transportation strategies, each characterised by its unique cost structure. The objective is to determine the most efficient routes and strategies to optimise the transportation flow. The concept of clustering routes is introduced into an MS supply chain

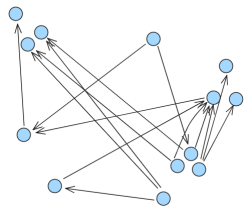


Figure 7.1: Routes of the supply chain

configuration.

Table 7.1 compares the supply chain configuration, transport modelling, and transport aggregation between the SCRC model proposal and those currently in the literature.

Table 7.1: Comparison of procurement transport optimisation models

Article	SC configuration	Decision level	Transportation mode	Transportation capacity constraint	Volume and weight optimisation	Source aggregation	Destination aggregation	Modelling framework
Schöneberg et al., 2010	MS	T	FTL|LTL	V		X		MILP
Schöneberg et al., 2013	MS	O	FTL|LTL	V		X		MILP
Díaz-Madroñero et al., 2017	MS	T	FTL|LTL|MR	V		X		MILP
Chitsaz et al., 2019	MS	O|T	MR	V		X		MILP
Baller et al., 2022	MS	O	MR|FTL|LTL|CES	V|W	X	X		MILP
Bou-Zeid et al., 2023	MR|FTL|LTL	T	MR	V				DA
SCRC	MM	T	FTL|LTL	V|W	X	X	X	MILP

Abbreviations. **T** (tactical), **O** Operational; **FTL** (full-truck load), **LTL** (less-than-truck load), **MR** (milk-run), **CES** (courier express service); **V** (volumetric), **W** (weight) **UH** Unit Holding; **DA** Decomposition Approach;

The main novelty of our proposal is the joint transport MM optimisation, i.e., between multiple sources and multiple destinations, while optimising the volume and weight mix using the FTL and LTL transport methods.

7.3 Problem Description

The objective of the proposed model is to reduce transportation costs in an MM supply chain by consolidating transport from different suppliers to various manufacturing plants from a tactical decision level. Figure 7.1 illustrates the considered supply chain, where circles are supply chain nodes and edges represent the established routes between them. Therefore, a node can be both a manufacturing facility and a supplier for other nodes, depending if the node has incoming and outcoming edges. Each route has a preset transport type, number of used trucks and volume and weight transported per week to procure the different manufacturing facilities. Depending on these parameters and route length, there is also a pre-established cost per week.

If a route is in low demand, i.e., the number of trucks needed to transport goods per week is less than one, then the route is normally delegated to a logistics company with higher shipping costs. Otherwise, the employed transport type is FTL. Efficiency rarely reaches 100% due to varying demands or the need for urgent sourcing, which necessitate more trucks or sending partially filled trucks. Moreover, the minimum required number

(a) Routes (b) Cluster

Figure 7.2: Clustered routes example

of trucks is determined by the route's total volume and weight, and the truck's maximum capacity in both volume and weight terms. Jointly managing the clustered routes offers the possibility of reducing the current costs. On the one hand, we eliminate the cost of delegating the routes in low demand because we consolidate all the routes in the cluster. On the other hand, we can reduce the number of trucks needed by mixing the routes dominated by weight and volume.

To optimise transportation decisions, we propose the concept of clusters by grouping routes for joint transportation management purposes. A cluster comprises both the receiving and sourcing nodes. We set both the receiving and sourcing central hubs nodes of the supply chain. The transport in the cluster can be separated into three stages: sourcing to the source central hub; transporting goods from the source central hub to the receiving central hub; distributing goods from the receiving central hub to the receiving nodes. The transportation cost associated with this cluster is derived in these three stages. Figure 7.2 illustrates the clustering process of a set of routes.

Additionally, each route has a set of routes to which it can be clustered. This is because some routes carry materials that are incompatible for joint transport. Furthermore, the beginning or end of routes could be in different countries, and routes are not mixed because of customs.

The objective of the proposed optimisation method is to find new clusters to minimise the transportation costs. Not all routes need to be clustered. Therefore, a solution is determined by the sets of clusters and non-clustered routes. A possible solution for the supply chain example is illustrated in Figure 7.3. Note that not all routes have been clustered.

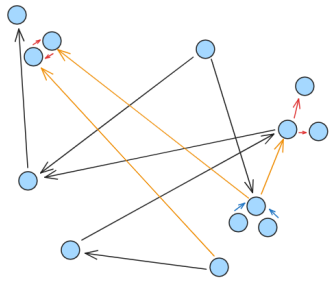


Figure 7.3: Clustered supply chain

The problem can be stated as follows:
Initial information:

- Transport details of routes (weight, volume, current cost).
- Distance among all the nodes of the supply chain.
- Set of edges that can be grouped together.
- Cost per kilometre.
- Maximum weight and volume allowed per truck.
- Parameters concerning consolidation efficiency.

To determine:

- Set of clusters for transport consolidation.
- Set of unaltered routes.

With constraints:

- Adhering to the association graph.

Objective:

- Minimise transportation costs.

7.4 Mathematical Formulation

In this section, we describe the developed SCRC model. The nomenclature is provided in Table 7.2

Table 7.2: Nomenclature of SCRC

Indices	
V	Set of vertices of the graph (v =1, 2,. . . , V)
E	Set of edges of the graphs that represent the current routes (e =1, 2,. . . ,E)
C	Set of edge cluster (e =1, 2,. . . ,C)
Set of Indices	

$\Gamma(e)$	Set of edges that can be grouped with e
$\delta^+(e)$	Receiving node of the edge
$\delta^-(e)$	Source node of the edge
Parameters	
v_e	Volume transported on route e
w_e	Weight transported on route e
d_{ij}	Distance between nodes $i \in V$ and $j \in V$
c_e	Initial cost of route on edge e
max_v	Maximum volume carried by a truck
max_w	Maximum weight carried by a truck
α	Efficiency of transport where volume dominates the route
β	Efficiency of transport where weight dominates the route
γ	Efficiency of transport between hubs
γ^+	Efficiency of transport between receiving hubs and receiving nodes
γ^-	Efficiency of transport between sourcing nodes and sourcing hub
Decision Variables	
Y_e	Binary variable indicating if e is not clustered
X^c	Binary variable indicating if cluster c is used
X_{ce}	Binary variable indicating if edge e belongs to cluster c
X^-_{ce}	Binary variable indicating if node $\delta^-(e)$ is the sourcing hub for the sourcing nodes in cluster c
X^+_{ce}	Binary variable indicating if node $\delta^+(e)$ is the receiving hub for the receiving nodes
$X^e_{ce'e''}$	Binary variable indicating if the edge e is in c and the source node of e' is the sourcing hub for outgoing goods and receiving node of e'' if the is the receiving hub for incoming goods
D_{ce}	Distance between the sourcing and receiving hub of cluster c if e is in the cluster, 0 if not.
D^-_{ce}	Distance from $\delta^-(e)$ to the sourcing hub if e is in cluster c
D^+_{ce}	Distance from $\delta^+(e)$ to the receiving hub if e is in cluster c
K_c	Transportation cost between sourcing hub and receiving hub at cluster c
K^-_c	Transportation cost between sourcing nodes and sourcing hub at cluster c
K^+_c	Transportation cost between receiving hub and receiving nodes at cluster c

The objective is to minimise transportation costs, i.e., the cost of the non-altered routes and the cost of the different clusters.

$$\min z = \sum_{e \in E} Y_e c_e + \sum_c \gamma K_c + \gamma^+ K_c^- + \gamma^- K_c \tag{7.1}$$

subject to:

$$Y_e + \sum_c X_{ce} = 1 \quad \forall e \in E \tag{7.2}$$

$$\sum_{e \in E} X_{ce} \leq M X^c \quad \forall c \in C \tag{7.3}$$

$$\sum_{e \in E} X_{ce} \geq 2X^c \quad \forall c \in C \tag{7.4}$$

$$\sum_{e \in E} X^-_{ce} = X^c \quad \forall c \in C \tag{7.5}$$

$$\sum_{e \in E} X^+_{ce} = X^c \quad \forall c \in C \tag{7.6}$$

$$\sum_{e' \in \Gamma(e)} X^+_{ce'} = X_{ce} \quad \forall c \in C \tag{7.7}$$

$$\sum_{e' \in \Gamma(e)} X^-_{ce'} = X_{ce} \quad \forall c \in C \tag{7.8}$$

$$\sum_{e', e'' \in \Gamma(e)} X^e_{ce'e''} = X_{ce} \quad \forall c \in C, e \in E \tag{7.9}$$

$$X^e_{ce'e''} + 2(1 - X_{ce}) + 1 \geq X^-_{ce'} + X^+_{ce''} \quad \forall c \in C, e \in E, e', e'' \in \Gamma(e) \tag{7.10}$$

$$\sum_{e' \in (E' \Gamma(e))} X_{ce'} \leq M(1 - X_{ce}) \quad \forall c \in C, e \in E \tag{7.11}$$

$$D^-_{ce} = \sum_{e' \in \Gamma(e)} X^-_{ce'} d_{\delta^-(e)\delta^-(e')} \quad \forall c \in C, e \in E \tag{7.12}$$

$$D^+_{ce} = \sum_{e' \in \Gamma(e)} X^-_{ce'} d_{\delta^+(e)\delta^+(e')} \quad \forall c \in C, e \in E \tag{7.13}$$

$$D_{ce} = \sum_{e', e'' \in \Gamma(e)} X^e_{ce'e''} d_{\delta^-(e')\delta^+(e'')} \quad \forall c \in C, e \in E \tag{7.14}$$

$$max_v K^-_{ce} \geq D^-_{ce} v_e \quad \forall c \in C, e \in E \tag{7.15}$$

$$max_w K^-_{ce} \geq D^-_{ce} w_e \quad \forall c \in C, e \in E \tag{7.16}$$

$$K^-_c = \sum_{e \in E} K^-_{ce} \quad \forall c \in C \tag{7.17}$$

$$max_v K^+_{ce} \geq D^+_{ce} v_e \quad \forall c \in C, e \in E \tag{7.18}$$

$$max_v K^+_{ce} \geq D^+_{ce} v_e \quad \forall c \in C, e \in E \tag{7.19}$$

$$K^+_c = \sum_{e \in E} K^+_{ce} \quad \forall c \in C \tag{7.20}$$

$$max_w K_c \geq (\sum_{e \in E} D_{ce} w_e)\alpha \quad \forall c \in C \tag{7.21}$$

$$max_v K_c \geq (\sum_{e \in E} D_{ce} v_e)\beta \quad \forall c \in C \tag{7.22}$$

Equation (7.2) determines if a route is clustered or not. If a cluster is formed, then it should have at least two routes (Equations (7.3-7.4)). For each created cluster, we should

set a sourcing hub, and a receiving hub (Equations (7.5-7.8)). Equation (7.9-7.10) links the sourcing hub and the receiving hub for all the routes in the cluster. Equations (7.11) guarantees that only compatible edges can be set in a cluster. For each clustered route, Equation (7.12) sets the distance from the outgoing node of the route to the sourcing hub. Equation (7.14) sets the distance from the sourcing hub to the receiving hub. Equation (7.13) establishes the distance from the receiving hub to the destination node of the route. Equations (7.15-7.16) determine the transportation cost from the outgoing node to the sourcing hub by taking into account the distance between them and the number of trucks needed, which depends on the volume and weight of the routes. Equation (7.17) determines the total transportation costs to the supplier hub. Similarly, Equations (7.18-7.20) determine the transportation costs of the cluster from the receiving hub to the destination nodes. Equations (7.21-7.22) determine the transportation cost from the sourcing hub to the receiving hub by taking into account the distance between them and the number of trucks needed, which depends on the volume and weight of routes and the efficiency parameters.

7.5 Branch and Price Solution Approach

There are two main resolution issues with the provided SCRC model. The first one is that a priori we do not know the exact number of needed clusters. Therefore, we could overestimate the number of clusters needed to find the optimal clustering of routes. The other is that the number of needed variables and constraints grows linearly in relation to the number of clusters set. Meanwhile, the decision space of the integer variables exponentially grows. This leads to an inefficient solution using current MILP solvers. Therefore, a column generation scheme based on branch and price is developed.

The key components of the branch and price algorithm are the master problem (MP), the restricted master problem (RMP), the pricing problem (PP) and the branching policy. The MP is a reformulation of a linear or integer programming model so that we can exploit the decomposable structure of the problem. However, the MP is typically too large to be solved directly because it involves a huge number of variables and columns that represent all the possible decision scenarios (e.g., all the possible routes in a routing problem or all the feasible clusters in our case). To manage this complexity, we consider only a smaller subset of variables, which leads to the formulation of the RMP to find new promising columns to be added to the RMP. The RMP only uses a subset of the variables and columns of the MP. Solving the RMP provides an optimum solution for the included subset of variables, but not necessarily for the entire set of original variables. Therefore, RMP integrality constraints are relaxed to find the current dual variables and new columns through the PP, also known as the subproblem, which is used to identify which variables (or columns) should be added to the RMP to potentially improve the current solution. The goal of the PP is to find variables with a negative reduced cost (in a minimisation problem) to be added to the RMP, which is likely to lead to a better objective value.

If there are no new columns with a negative reduced cost, then we have optimally solved the integrality relaxed MP. If all the variables associated with the solution are integers, then we have found an integer solution to the current branch. If not, we have to branch the current branch by adding further constraints to find integer solutions through the relaxed RMP. Figure 7.4 exmplifies the Branch & Price Algorithm.

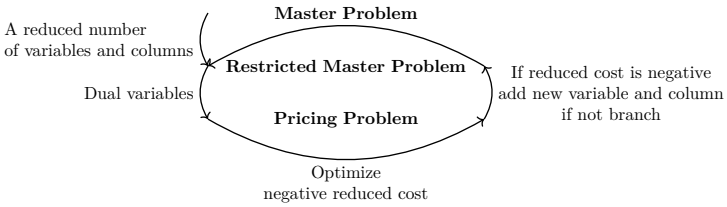


Figure 7.4: Branch and Price

7.5.1 Formulation of the Master Problem and the Reduced Master Problem

Here we firstly reformulate the model as a set-partitioning formulation (Dantzig–Wolfe decomposition). The key idea of the set-partitioning formulation for the SCRC is to enumerate all the feasible clusters with their associated costs. Let P be the set of all feasible clusters that satisfy the constraints imposed by the SCRC model. For each cluster $p \in P$, a binary variable λ_p is defined that equals to 1 if cluster p appears in the solution and 0 otherwise. Binary parameter a_e^p equals to 1 if cluster p contains the edge $e \in E$. Parameter c_p represents the cost associated with cluster c. In this case, we consider that the pre-established routes are clusters with pre-established cost and with $a_e^p = 1$ if e is the current route, and 0 otherwise. Therefore, the MP of our problem is:

$$\min \sum_{p \in P} c_p \lambda_p \tag{7.23}$$

subject to:

$$\sum a_e^p \lambda^p = 1 \quad \forall e \in E \tag{7.24}$$

$$\lambda_p \in \{0, 1\} \quad \forall p \in P \tag{7.25}$$

The objective function 7.23 is to minimise the cost of the clusters needed to transport goods. Meanwhile, 7.24 ensures that each route is only used in one cluster.

Here we employ the initial routes as the initial clusters used in the RMP with their associated costs.

$$\min \sum_{p \in P} c_p \lambda_p \tag{7.26}$$

subject to:

$$\sum a_e^p \lambda^p = 1 \quad \forall e \in E \tag{7.27}$$

$$0 \leq \lambda_p \leq 1 \quad \forall p \in P \tag{7.28}$$

Dual values are obtained by solving the RMP. Let $\pi_e, e \in E$, be dual variables associated with constraints (7.27). Then the reduced cost associated with cluster $p \in P$ is $c_p - \sum_e a_e^p \pi_e$. The feasible cluster with a negative reduced cost can improve the current solution to the RMP. For each iteration, we solve the subproblem that consists of finding a feasible cluster

with the minimum negative reduced cost. The value of the variables associated with negative reduced cost (expressed as new columns) are added to the RMP. Then the RMP is solved to generate a new set of dual values. This process is repeated until no further variables with a negative reduced cost are discovered.

7.5.2 Pricing Subproblem

We solve the pricing problem using an MILP approach. The indices and parameters are the same as the model proposed before, but the decisions variables do not depend on the indices of clusters. Table 7.3 describes the new decision variables.

Table 7.3: Nomenclature of the pricing subproblem

Decision Variables	
X_e	Binary variable indicating if edge e belongs to the new cluster
X_e^-	Binary variable indicating if $\delta^-(e)$ is the sourcing hub for the suppliers in the new cluster
X_e^+	Binary variable indicating if $\delta^+(e)$ is the receiving hub for the nodes in the new cluster
$X^e_{e'e''}$	Binary variable indicating if edge e is in c and the source node of e' is the sourcing hub for outgoing goods and receiving node of e'' is receiving hub for incoming goods in the new cluster
D_e	Distance between the sourcing and receiving hub of the new cluster if e is in the cluster, 0 if not.
D_e^-	Distance from $\delta^-(e)$ to the sourcing hub if e is in the new cluster, 0 if not
D_e^+	Distance from $\delta^+(e)$ to the receiving hub if e is in the new cluster, 0 if not
K	Transportation cost between the sourcing nodes and the sourcing hub in the new cluster
K^-	Transportation cost between the sourcing hub and the receiving hub in the new cluster
K^+	Transportation cost between the receiving hub in the new cluster

The objective is to find new clusters with negative reduced costs. That is:

$$\min z = \gamma K + \gamma^- K^- + \gamma^+ K^+ - \sum_{e \in E} \pi_e * X_e \tag{7.29}$$

subject to:

$$\sum_{e \in E} X_e \geq 2 \tag{7.30}$$

$$\sum_{e \in E} X_e^- = 1 \tag{7.31}$$

$$\sum_{e \in E} X_e^+ = 1 \tag{7.32}$$

$$\sum_{e' \in \Gamma(e)} X_{e'}^+ = X_e \quad \forall e \in E \tag{7.33}$$

$$\sum_{e' \in \Gamma(e)} X_{e'}^- = X_e \quad \forall e \in E \tag{7.34}$$

$$\sum_{e', e'' \in \Gamma(e)} X_{e'e''}^e = X_e \quad \forall e \in E \tag{7.35}$$

$$X_{e'e''}^e + 2(1 - X_e) + 1 \geq X_{e'}^- + X_{e''}^+ \quad \forall e \in E, e', e'' \in \Gamma(e) \tag{7.36}$$

$$\sum_{e' \in E' \Gamma(e)} X_{e'} \leq M(1 - X_e) \quad \forall e \in E \tag{7.37}$$

$$D_e^- = \sum_{e' \in \Gamma(e)} X_{e'}^- d_{\delta^-(e)\delta^-(e')} \quad \forall e \in E \tag{7.38}$$

$$D_e^+ = \sum_{e' \in \Gamma(e)} X_{e'}^- d_{\delta^+(e)\delta^+(e')} \quad \forall e \in E \tag{7.39}$$

$$D_e = \sum_{e', e'' \in \Gamma(e)} X_{ce'e''}^e d_{\delta^-(e')\delta^+(e'')} \quad \forall e \in E \tag{7.40}$$

$$max_v K_e^- \geq D_e^- v_e \quad \forall e \in E \tag{7.41}$$

$$max_w K_e^- \geq D_e^- w_e \quad \forall e \in E \tag{7.42}$$

$$K^- = \sum_{e \in E} K_e^- \tag{7.43}$$

$$max_v K_e^+ \geq D_e^+ v_e \quad \forall e \in E \tag{7.44}$$

$$max_v K_e^+ \geq D_e^+ v_e \quad \forall e \in E \tag{7.45}$$

$$K^+ = \sum_{e \in E} K_e^+ \tag{7.46}$$

$$max_w K \geq (\sum_{e \in E} D_e w_e)\alpha \tag{7.47}$$

$$max_v K \geq (\sum_{e \in E} D_e v_e)\beta \tag{7.48}$$

Similarly, (7.30-7.48) restrict the clusters used in the pricing problem to feasible ones.

Table 7.4: Nomenclature for instance types

Size	Number of Routes
Small	(0, 500]
Medium	(500,1000]
Large	(1000,1500]
Extra Large	(1500-2500]

7.5.3 Branching

Here, we use the depth-first branching rule, which once we can not find any new promising solutions through pricing problem we set the variable with the largest fractional to one. That is, we will fix a cluster that must be used. After fixing the variables, more columns are generated using the pricing algorithm. Repeating this branching strategy leads to an integer solution of the RMP. In our case, this method of branching does not guarantee to find an optimal solution, but it does guarantee a feasible one.

7.6 Computational Experiments

Here we compare both the performance of the SCRC model and the developed price and branch approach. For this purpose, 200 instances of different sizes were artificially generated and inspired in a first-tier supplier of the automotive sector. Depending on the total number of routes, we grouped instances into four main types: small, medium, large and extra larges. The routes in the instances were generated by randomly assigning a source (supplier) and a destination (manufacturing facility) of the supply chain under study. Furthermore, each route was randomly assigned to be constrained by volume or weight.

We used the Julia programming language and the JuMP package to run all the calculations. To solve both the RMP and PP, we have used the Gurobi commercial solver.

The Table 7.4 illustrates how we have grouped the instances.

Table 7.5 presents the mean objective value and resolution duration aggregated by the size of the instace for the different resolution procedures. For the SCRC MILP model we have set a time limit of 2 hours. Meanwhile, for the branch and price resolution procudere, we have set a time limit of 30 minutes for each node iteration.

As we can observe, the branch and price algorithm outperforms the MILP model in both terms of solution quality and resolution time.

7.7 SCRC As A Decision Support Tool

The SCRC model can be used as a decision tool for clustering routes of various supply chains. Through the different efficiency parameters of the algorithm, decision makers can analyse the various clustering routes proposed by the algorithm. If we set all the efficiency parameters to 1, then we consider that clustering will not have any inefficiencies, which is rarely the case. Once we have identified where inefficiencies are from, for example, the sourcing to the main hub, then we can adjust this parameter and see how the system behaves. Figure illustrates the different savings proposed by algorithms when we increase the inefficiency of one of the parameters. Furthermore, if a decision maker thinks that a proposed cluster is not well suited for the supply chain being considered, additional constraints can be set in

Table 7.5: Mean objective value and resolution duration per resolution procedure

Type	Resolution pocedure	Transport Cost	Duration
Extra large	B&P	6864750.622806	1826 sec
Extra large	MILP	7507608.574584	11520 sec
Large	B&P	3811390.489451	1803 sec
Large	MILP	4498435.223205	8820 sec
Medium	B&P	2048727.759246	1435 sec
Medium	MILP	2531413.865415	8010 sec
Small	B&P	622577.984265	43 sec
Small	MILP	673251.233579	5825 sec

the mathematical model by selecting which routes can a route be clustered.

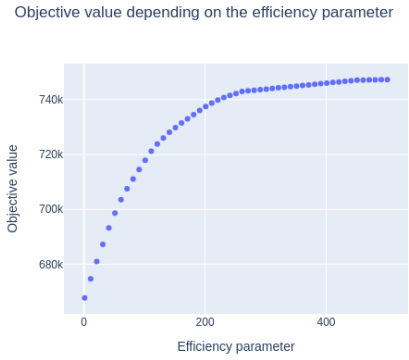


Figure 7.5: Transport cost vs. Efficiency parameter

As we can see in Figure 7.5, as the efficiency parameter increases the transport cost also increases. Furthermore, the transport cost can not overpass the initial cost of the problem, since if the new clusters are too costly it will choose to use the initial configuration.

SCRC model has been used as a decision tool for clustering routes at a first tier supplier of the automotive sector.

7.8 Conclusions and Future Research

In conclusion, this paper presents a pioneering effort in route clustering for a supply chain with multiple suppliers and manufacturing facilities. It introduces a mathematical model designed to consolidate the transport resource of large supply chains. Furthermore, the model has different parameters that the decision maker must decide. Not only the development of a mathematical model, but also the subsequent application of a branch and priced method to ascertain near-optimal solutions, underscore the robustness and innovative approach of this study. The decision tool, validated in a first-tier supplier of the automotive industry, provides a solid foundation for future explorations and applications in procurement transport planning.

Looking ahead, future research alternatives have been identified, and promise to further enhance the model's utility and applicability. These include exploring alternative metrics for evaluating the transport cost of each cluster by, for example, adding a vehicle routing problem for sourcing and delivery to the central hubs. Furthermore, it is possible to integrate the model to make operational decisions to dynamically adjust routes based on inventory levels and the demand of the manufacturing facilities. Additionally, it offers the possibility of addressing transport decisions about using reusable packaging in the supply. Regarding the resolution procedure, a future research potential is identified. For instance, using other branching strategies to ensure optimal solutions for medium-sized instances and applying metaheuristcs for the pricing problem for larger instances.

Chapter 8

Global Results

Comparing Article 2 and 3.

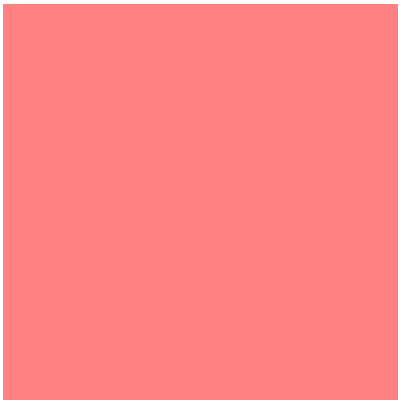


Figure 8.1: Cross-chapter comparison

Chapter 9

Conclusions

Final thoughts.

Bibliography

Achamrah, F. E., Riane, F., Sahin, E., & Limbourg, S. (2022). An artificial-immune-system-based algorithm enhanced with deep reinforcement learning for solving returnable transport item problems. *Sustainability*, *14*(10), 5805. https://doi.org/10.3390/su14105805

Ali, S., Ramos, A. G., Carravilla, M. A., & Oliveira, J. F. (2022). On-line three-dimensional packing problems: A review of off-line and on-line solution approaches. *Computers and Industrial Engineering*, *168*. https://doi.org/10.1016/j.cie.2022.108122

Alonso, M. T., Alvarez-Valdes, R., & Parreño, F. (2020). A grasp algorithm for multi container loading problems with practical constraints. *4OR*, *18*(1), 49–72. https://doi.org/10.1007/s10288-018-0397-z

Alonso, M. T., Martinez-Sykora, A., Alvarez-Valdes, R., & Parreño, F. (2022). The pallet-loading vehicle routing problem with stability constraints: The pallet-loading vehicle routing problem with stability constraints. second revision. *European Journal of Operational Research*, *302*(3), 860–873. https://doi.org/10.1016/j.ejor.2022.01.035

Baller, R., Fontaine, P., Minner, S., & Lai, Z. (2022). Optimizing automotive inbound logistics: A mixed-integer linear programming approach. *Transportation Research Part E: Logistics and Transportation Review*, *163*. https://doi.org/10.1016/j.tre.2022.102734

Berman, O., & Wang, Q. (2006). Inbound logistic planning: Minimizing transportation and inventory cost. *Transportation Science*, *40*(3), 287–299. https://doi.org/10.1287/trsc.1050.0130

Bortfeldt, A., & Wäscher, G. (2013). Constraints in container loading-a state-of-the-art review. *European Journal of Operational Research*, *229*(1), 1–20. https://doi.org/10.1016/j.ejor.2012.12.006

Bou-Zeid, I., Alpan, G., Penz, B., Serrano, C., Nguyen, A., & Khatouf, M.-A. (2023). *Optimal Multi-Period Tactical Transportation Plan for Inbound Logistics: A case study in the automotive industry. Optimal Multi-Period Tactical Transportation Plan for Inbound Logistics: A case study in the automotive industry.* (tech. rep.).

Chen, C. S., Lee, S. M., & Shen, Q. S. (1995). *European journal of operational research an analytical model for the container loading problem* (tech. rep.).

Chen, M., Huo, J., & Duan, Y. (2023). A hybrid biogeography-based optimization algorithm for three-dimensional bin size designing and packing problem. *Computers and Industrial Engineering*, *180*. https://doi.org/10.1016/j.cie.2023.109239

Chitsaz, M., Cordeau, J. F., & Jans, R. (2019). A unified decomposition matheuristic for assembly, production, and inventory routing. *INFORMS Journal on Computing*, *31*(1), 134–152. https://doi.org/10.1287/ijoc.2018.0817

Chopra, S., & Meindl, P. (2015). *Supply chain management : Strategy, planning, and operation.*

Dantzig, G. B., & Wolfe, P. (1960). Decomposition Principle for Linear Programs. *Operations Research*, *8*(1), 101–111.

Desrosiers, J., & Lübbecke, M. E. (2011, January). Branch-Price-and-Cut Algorithms. In *Wiley encyclopedia of operations research and management science*. Wiley. https://doi.org/10.1002/9780470400531.eorms0118

Díaz-Madroñero, M., Mula, J., & Peidro, D. (2014, September). A review of discrete-time optimization models for tactical production planning. https://doi.org/10.1080/00207543.2014.899721

Díaz-Madroñero, M., Mula, J., & Peidro, D. (2017). A mathematical programming model for integrating production and procurement transport decisions. *Applied Mathematical Modelling*, *52*, 527–543. https://doi.org/10.1016/j.apm.2017.08.009

Díaz-Madroñero, M., Peidro, D., & Mula, J. (2014). A fuzzy optimization approach for procurement transport operational planning in an automobile supply chain. *Applied Mathematical Modelling*, *38*(23), 5705–5725. https://doi.org/10.1016/j.apm.2014.04.053

Díaz-Madroñero, M., Peidro, D., & Mula, J. (2015, October). A review of tactical optimization models for integrated production and transport routing planning decisions. https://doi.org/10.1016/j.cie.2015.06.010

Dominguez, O., Juan, A. A., Barrios, B., Faulin, J., & Agustin, A. (2016). Using biased randomization for solving the two-dimensional loading vehicle routing problem with heterogeneous fleet. *Annals of Operations Research*, *236*(2), 383–404. https://doi.org/10.1007/s10479-014-1551-4

Dyckhoff, H. (1990). *A typology of cutting and packing problems* (tech. rep.).

Fleischmann, B., Kopfer, H., & Sürie, C. (2015). Transport planning for procurement and distribution. In H. Stadtler, C. Kilger, & H. Meyr (Eds.), *Sup-*

ply chain management and advanced planning (pp. 225–240). Springer. https://doi.org/10.1007/978-3-642-55309-7

Ford, L. R., & Fulkerson, D. R. (1958). Constructing Maximal Dynamic Flows from Static Flows. *Operations Research*, *6*(3), 419–433.

Gilmore, P. C., & Gomory, R. E. (1961). A Linear Programming Approach to the Cutting-Stock Problem. *Operations Research*, *9*, 849–859.

Glock, C. H. (2017). Decision support models for managing returnable transport items in supply chains: A systematic literature review. *International Journal of Production Economics*, *183*, 561–569. https://doi.org/10.1016/j.ijpe.2016.02.015

Gonçalves, J. F., & Resende, M. G. (2013). A biased random key genetic algorithm for 2d and 3d bin packing problems. *International Journal of Production Economics*, *145*(2), 500–510. https://doi.org/10.1016/j.ijpe.2013.04.019

Govindan, K., Soleimani, H., & Kannan, D. (2015). Reverse logistics and closed-loop supply chain: A comprehensive review to explore the future. *European Journal of Operational Research*, *240*(3), 603–626. https://doi.org/10.1016/j.ejor.2014.07.012

Guo, M., Chan, H. K., Kong, X. T., & Thadani, D. R. (2024). Incentivising-by-penalty: The optimal return strategy for a reusable transport item rental platform. *Transportation Research Part D: Transport and Environment*, *134*, 104339. https://doi.org/https://doi.org/10.1016/j.trd.2024.104339

Harrath, Y. (2022). A three-stage layer-based heuristic to solve the 3d bin-packing problem under balancing constraint. *Journal of King Saud University - Computer and Information Sciences*, *34*, 6425–6431. https://doi.org/10.1016/j.jksuci.2021.07.007

Hashemi-Amiri, O., Ghorbani, F., & Ji, R. (2023). Integrated supplier selection, scheduling, and routing problem for perishable product supply chain: A distributionally robust approach. *Computers and Industrial Engineering*, *175*. https://doi.org/10.1016/j.cie.2022.108845

Hein, F., & Almeder, C. (2016). Quantitative insights into the integrated supply vehicle routing and production planning problem. *International Journal of Production Economics*, *177*, 66–76. https://doi.org/10.1016/j.ijpe.2016.04.014

Hellström, D., & Johansson, O. (2010). The impact of control strategies on the management of returnable transport items. *Transportation Research Part E: Logistics and Transportation Review*, *46*(6), 1128–1139. https://doi.org/10.1016/j.tre.2010.05.004

Iassinovskaia, G., Limbourg, S., & Riane, F. (2017). The inventory-routing problem of returnable transport items with time windows and simultaneous pickup and delivery in closed-loop supply chains. *International Journal of Production Economics*, *183*, 570–582. https://doi.org/10.1016/j.ijpe.2016.06.024

Junqueira, L., Morabito, R., & Sato Yamashita, D. (2012a). Mip-based approaches for the container loading problem with multi-drop constraints. *Annals of Operations Research*, *199*(1), 51–75. https://doi.org/10.1007/s10479-011-0942-z

Junqueira, L., Morabito, R., & Sato Yamashita, D. (2012b). Three-dimensional container loading models with cargo stability and load bearing constraints. *Computers and Operations Research*, *39*(1), 74–85. https://doi.org/10.1016/j.cor.2010.07.017

Junqueira, L., Oliveira, J. F., Carravilla, M. A., & Morabito, R. (2013). An optimization model for the vehicle routing problem with practical three-dimensional loading constraints. *International Transactions in Operational Research*, *20*(5), 645–666. https://doi.org/10.1111/j.1475-3995.2012.00872.x

Kurpel, D. V., Scarpin, C. T., Pécora Junior, J. E., Schenekemberg, C. M., & Coelho, L. C. (2020). The exact solutions of several types of container loading problems. *European Journal of Operational Research*, *284*(1), 87–107. https://doi.org/10.1016/j.ejor.2019.12.012

Langley, C. J., Novack, R. A., Gibson, G., & Coyle, J. H. (2020). *Supply chain management: A logistics perspective* (11th). Cengage Learning.

Li, Y., Chen, M., & Huo, J. (2022). A hybrid adaptive large neighborhood search algorithm for the large-scale heterogeneous container loading problem. *Expert Systems with Applications*, *189*. https://doi.org/10.1016/j.eswa.2021.115909

Liu, G., Li, L., Chen, J., & Ma, F. (2020). Inventory sharing strategy and optimization for reusable transport items. *International Journal of Production Economics*, *228*, 107742. https://doi.org/10.1016/j.ijpe.2020.107742

Lübbecke, M. E., & Desrosiers, J. (2005). Selected Topics in Column Generation. *Operations Research*, *53*(6), 1007–1023.

Mahmoudi, M., & Parviziomran, I. (2020). Reusable packaging in supply chains: A review of environmental and economic impacts, logistics system designs, and operations management. *International Journal of Production Economics*, *228*, 107730. https://doi.org/10.1016/j.ijpe.2020.107730

Mahvash, B., Awasthi, A., & Chauhan, S. (2018). A column generation-based heuristic for the three-dimensional bin packing problem with rotation. *Journal of the Operational Research Society*, *69*, 78–90. https://doi.org/10.1057/s41274-017-0186-7

Meyer, A., & Amberg, B. (2018). Transport concept selection considering supplier milk runs – an integrated model and a case study from the automotive industry. *Transportation Research Part E: Logistics and Transportation Review*, *113*, 147–169. https://doi.org/10.1016/j.tre.2017.07.004

Mosca, A., Vidyarthi, N., & Satir, A. (2019, January). Integrated transportation – inventory models: A review. https://doi.org/10.1016/j.orp.2019.100101

Mula, J., Peidro, D., Díaz-Madroñero, M., & Vicens, E. (2010). Mathematical programming models for supply chain production and transport planning. *European Journal of Operational Research*, *204*(3), 377–390. https://doi.org/10.1016/j.ejor.2009.09.008

Paquay, C., Schyns, M., & Limbourg, S. (2016). International transactions in operational research a mixed integer programming formulation for the three-dimensional bin packing problem deriving from an air cargo application. *Intl. Trans. in Op. Res*, *23*, 187–213. https://doi.org/10.1111/itor.12111

Que, Q., Yang, F., & Zhang, D. (2023). Solving 3d packing problem using transformer network and reinforcement learning. *Expert Systems with Applications*, *214*. https://doi.org/10.1016/j.eswa.2022.119153

Sancak, E., & Salman, F. S. (2011). Multi-item dynamic lot-sizing with delayed transportation policy. *International Journal of Production Economics*, *131*(2), 595–603. https://doi.org/10.1016/j.ijpe.2011.02.004

Sarkar, B., Ullah, M., & Kim, N. (2017). Environmental and economic assessment of closed-loop supply chain with remanufacturing and returnable transport items. *Computers & Industrial Engineering*, *111*, 148–163. https://doi.org/10.1016/j.cie.2017.07.003

Schöneberg, T., Koberstein, A., & Suhl, L. (2010). An optimization model for automated selection of economic and ecologic delivery profiles in area forwarding based inbound logistics networks. *Flexible Services and Manufacturing Journal*, *22*(3-4), 214–235. https://doi.org/10.1007/s10696-011-9084-5

Schöneberg, T., Koberstein, A., & Suhl, L. (2013). A stochastic programming approach to determine robust delivery profiles in area forwarding inbound logistics networks. *OR Spectrum*, *35*(4), 807–834. https://doi.org/10.1007/s00291-013-0349-0

Silva, E. F., Toffolo, T. A. M., & Wauters, T. (2019). Exact methods for three-dimensional cutting and packing: A comparative study concerning single container problems. *Computers and Operations Research*, *109*, 12–27. https://doi.org/10.1016/j.cor.2019.04.020

Smith, A. (2023). Review of everything. *J. Rev.*

Soysal, M., Koç, Ç., Çimen, M., & İbiş, M. (2023a). Managing returnable transport items in a vendor managed inventory system. *Socio-Economic Planning Sciences*, *86*, 101504. https://doi.org/10.1016/j.seps.2022.101504

Soysal, M., Koç, Ç., Çimen, M., & İbiş, M. (2023b). Managing returnable transport items in a vendor managed inventory system. *Socio-Economic Planning Sciences*, *86*, 101504. https://doi.org/https://doi.org/10.1016/j.seps.2022.101504

Stadtler, H., Kilger, C., & Meyr, H. (2015). *Supply chain management and advanced planning* (Vol. 5th Edition).

Tresca, G., Cavone, G., Carli, R., Cerviotti, A., & Dotoli, M. (2022). Automating bin packing: A layer building matheuristics for cost effective logistics. *IEEE Transactions on Automation Science and Engineering*, *19*, 1599–1613. https://doi.org/10.1109/TASE.2022.3177422

Wäscher, G., Haußner, H., & Schumann, H. (2007). An improved typology of cutting and packing problems. *European Journal of Operational Research*, *183*(3), 1109–1130. https://doi.org/10.1016/j.ejor.2005.12.047

Wei, L., Zhang, Z., Zhang, D., & Lim, A. (2015). A variable neighborhood search for the capacitated vehicle routing problem with two-dimensional loading constraints. *European Journal of Operational Research*, *243*(3), 798–814. https://doi.org/10.1016/j.ejor.2014.12.048

Zhang, Q., Segerstedt, A., Tsao, Y., & Liu, B. (2015). Returnable packaging management in automotive parts logistics: Dedicated mode and shared mode. *International Journal of Production Economics*, *168*, 234–244. https://doi.org/10.1016/j.ijpe.2015.07.002

Zhang, Y., Che, A., & Chu, F. (2022). Improved model and efficient method for bi-objective closed-loop food supply chain problem with returnable transport items. *International Journal of Production Research*, *60*(3), 1051–1068. https://doi.org/10.1080/00207543.2020.1851057

Zhang, Y., Chu, F., Che, A., & and, Y. L. (2024). Closed-loop inventory routing problem for perishable food with returnable transport items selection. *International Journal of Production Research*, *62*(1-2), 501–521. https://doi.org/10.1080/00207543.2023.2275639

Zhao, H., Zhu, C., & Huang, H. (2022). Learning practically feasible policies for online 3d bin packing. *65*(112105), 17. https://doi.org/10.1007/s11432-021-3348-6

Zhao, X., Bennell, J. A., Bektaşa, T., Bektaşa, B., & Dowsland, K. (2016). International transactions in operational research a comparative review of 3d container loading algorithms. *Intl. Trans. in Op. Res*, *23*, 287–320. https://doi.org/10.1111/itor.12094

Zhu, W., & Lim, A. (2012). A new iterative-doubling greedy-lookahead algorithm for the single container loading problem. *European Journal of Operational Research*, *222*(3), 408–417. https://doi.org/10.1016/j.ejor.2012.04.036

Appendix A

Mathematical Model

Table A.1: Nomenclature of Supply Chain Operation Transport Planning

Sets of indices	
I:	Set of products $(i = 1, 2, \ldots, I)$
J :	Set of groups composed of products that must be ordered together $(j = 1, 2, \ldots, J)$
K :	Set of trucks $(k = 1, 2, \ldots, K)$
T :	Set of planning periods (days) $(t = 1, 2, \ldots, T)$
Decision variables	
$Q_{ikt:}$	Units transported of i by k in period t (units)
G_{ijkt} :	Units transported of i corresponding to group j by k in period t (units)
C_{kt} :	Volume of containers transported by k in period t (units)
I_{it} :	Inventory amount of i at the end of period t (units)
$K_{jkt:}$	Number of lots to order of products of group j by k in period t
Y_{kt}	Binary variable indicating whether a truck k has been used in period t
Objective functions	
z_1 :	Total number of trucks utilized
z_2	Total inventory amount generated
Parameters	
u_i :	Amount of product i that fits in a container (units)
l_j :	Number of units of each group lot j (units)
b_{ij}	1 if product i belongs to group j, and 0 otherwise
D_{it} :	Demand of product i in t (units) (considered firm)
Γ_k :	Volume of truck k
Γ_i :	Volume of container carrying product i
m :	Minimum truck occupation (in containers)
$I0_i$	Inventory amount of i in period 0
IC_i	Inventory cost of i
C_k	Cost of using truck k
SC_i	Number of days of stock coverage for product i

Minimize the total cost transport and inventory

$$\text{Min}\, z \cong \sum_{k=1}^{K}\sum_{t=1}^{T} Y_{kt} + \sum_{i=1}^{I}\sum_{t=1}^{T} I_{it} \cdot IC_i$$

subject to

$$\begin{aligned}
&I_{it} = I_{i(t-1)} - D_{it} + \sum_{k=1}^{K} Q_{ikt} \quad \forall i \in I, t \in T,\\
&Q_{ikt} = \sum_{j=1}^{J} G_{ijkt} \quad \forall i \in I, k \in K, t \in T,\\
&G_{ijkt} = K_{jkt} \cdot l_j \cdot b_{ij} \quad \forall i \in I, j \in I, k \in K, t \in T,\\
&C_{kt} = \sum_{i=1}^{l} Q_{ikt}/u_i \cdot \Gamma_i \quad \forall k \in K, t \in T,\\
&C_{kt} \leqslant \Gamma_k \cdot Y_{kt} \quad \forall k \in K, t \in T,\\
&C_{kt} \geqslant m \cdot \Gamma_k Y_{kt} \quad \forall k \in K, \in T\\
&I_{nt} \geq \begin{cases} \sum_{s\in\{1,\ldots,SC_i\}} D_{i,t+s} & \text{if } i \in I, t \in \{1,\ldots,|T| - SC_i\}\\ \sum_{s\in\{1,\ldots,SC_i\}} D_{i,|T|-SC_i+s} & \text{other wise}\end{cases}\\
&Y_{kt} \leqslant 1 \quad \forall k, t,\\
&I_{it}, Q_{ikt}, G_{ijkt}, C_{kt}, K_{jkt}, Y_{kt} \geqslant 0 \text{ integer} \quad \forall i \in I, j \in I, k \in K, t \in T.
\end{aligned}$$

In our case $|J| = |I|$, one group for each product.

Appendix B

Full Non Linear Model

Table B.1: Explicit binary variables replacing aggregation conventions

Variable	**Type**	**Description**
YF_{ijmr}	Binary	1 if any product p uses full RTIs on route (i, j) by mode m, type r; 0 otherwise.
YI_{ijmr}	Binary	1 if any product p uses inlay RTIs on route (i, j) by mode m, type r; 0 otherwise.
YM_{ijm}	Binary	1 if mode m is active on route (i, j) for any RTI type r; 0 otherwise.
$YEOUT_{ir}$	Binary	1 if node i dispatches empty RTIs of type r to any destination j by any mode m; 0 otherwise.
$YEIN_{ir}$	Binary	1 if node i receives empty RTIs of type r from any origin j by any mode m; 0 otherwise.

Objective function.

$$\min \quad \sum_{(i,j)\in E} \sum_{m\in\mathcal{M}} TC_{ijm} \; + \; \sum_{r\in R} p_r^{\mathrm{RTI}} \cdot P_r \; + \; \sum_{i\in H} CH_i \tag{B.1}$$

Flow conservation.

$$\sum_{j \in V} \sum_{m \in \mathcal{M}} N_{ijmr} = \sum_{j \in V} \sum_{m \in \mathcal{M}} N_{jimr} \qquad \forall\, i \in V,\ r \in R \tag{B.2}$$

Flow–binary consistency.

$$NF_{ijmpr} \leq M \cdot XF_{ijmpr}, \quad XF_{ijmpr} \leq NF_{ijmpr} \cdot bf_{ijmpr} \quad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R \tag{B.3}$$

$$NI_{ijmpr} \leq M \cdot XI_{ijmpr}, \quad XI_{ijmpr} \leq NI_{ijmpr} \cdot bf_{jimpr} \quad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R \tag{B.4}$$

$$NE_{ijmr} \leq M \cdot XE_{ijmr}, \quad XE_{ijmr} \leq NE_{ijmr} \cdot be_{ijmr} \quad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.5}$$

$$N_{ijmr} = \sum_{p \in P} NF_{ijmpr} + \sum_{p \in P} NI_{ijmpr} + NE_{ijmr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.6}$$

Route activation.

$$3 \cdot X_{ijmr} \geq YF_{ijmr} + XE_{ijmr} + YI_{ijmr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.7}$$

$$X_{ijmr} \leq YF_{ijmr} + XE_{ijmr} + YI_{ijmr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.8}$$

Full-route activation.

$$\sum_{m \in \mathcal{M}} \sum_{r \in R} XF_{ijmpr} = bp_{ijp} \qquad \forall\, (i,j) \in E,\ p \in P \tag{B.9}$$

Transport mode selection.

$$YM_{ijm} \leq bm_{ijm} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M} \tag{B.10}$$

$$\sum_{m \in \mathcal{M}} YM_{ijm} \leq 1 \qquad \forall\, (i,j) \in E \tag{B.11}$$

Hub nodes.

$$YEOUT_{ir} + YEIN_{ir} \leq 1 \qquad \forall\, i \in V \backslash H, r \in R \tag{B.12}$$

$$CH_i = fhc_i \cdot XH_i + vhc_i \sum_{j \in V} \sum_{m \in \mathcal{M}} \sum_{r \in R} V_{ijmr} \qquad \forall\, i \in H \tag{B.13}$$

Demand satisfaction.

$$NF_{ijmpr} = \mu_{ijmpr} \cdot XF_{ijmpr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R \tag{B.14}$$

Inlay return flows.

$$NI_{jimpr} = ric_{ijpr} \cdot \mu_{ijmpr} \cdot XF_{ijmpr} \qquad \forall\, (i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \in R \tag{B.15}$$

Daily average volumes.

$$VF_{ijmr} = \left(\sum_{p \in P} NF_{ijmpr}\right) v^f \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.16}$$

$$VI_{ijmr} = \left(\sum_{p \in P} NI_{ijmpr}\right) v^f \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.17}$$

$$VE_{ijmr} = NE_{ijmr} \cdot v^e \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.18}$$

$$V_{ijmr} = VF_{ijmr} + VI_{ijmr} + VE_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.19}$$

Per-shipment RTI counts.

$$NSF_{ijmr} = \left(\sum_{p \in P} NF_{ijmpr}\right) q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.20}$$

$$NSI_{ijmr} = \left(\sum_{p \in P} NI_{ijmpr}\right) q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.21}$$

$$NSE_{ijmr} = NE_{ijmr} \cdot q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.22}$$

$$NS_{ijmr} = NSF_{ijmr} + NSI_{ijmr} + NSE_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.23}$$

Per-shipment volumes.

$$VSF_{ijmr} = VF_{ijmr} \cdot q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.24}$$

$$VSI_{ijmr} = VI_{ijmr} \cdot q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.25}$$

$$VSE_{ijmr} = VE_{ijmr} \cdot q_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.26}$$

$$VS_{ijmr} = VSF_{ijmr} + VSI_{ijmr} + VSE_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ r \in R \tag{B.27}$$

Queue time bounds.

$$q_m^{\min} \cdot YM_{ijm} \ \le\ q_{ijm} \ \le\ q_m^{\max} \cdot YM_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M} \tag{B.28}$$

Shipment volume bounds.

$$\omega_{ijm} \cdot YM_{ijm} \ \le\ \sum_{r \in R} VS_{ijmr} \ \le\ \delta_{ijm} \cdot YM_{ijm} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M} \tag{B.29}$$

Total RTI pool requirements.

$$TNCF_{ijmr} = (1+\alpha^f)\left(NSF_{ijmr} + l_{ijm} \cdot \sum_{p \in P} NF_{ijmpr} + XF_{ijmr}\big(ssf^i_{ijpr} + ssf^j_{ijpr}\big)\right) \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \tag{B.30}$$

$$TNCE_{ijmr} = (1+\alpha^e)\Big(NSE_{ijmr} + l_{ijm} \cdot NE_{ijmr} + XE_{ijmr}\big(sse^i_{ijpr} + sse^j_{ijpr}\big)\Big) \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \tag{B.31}$$

$$TNCI_{ijmr} = (1+\alpha^f)\left(NSI_{ijmr} + l_{ijm} \cdot \sum_{p \in P} NI_{ijmpr}\right) \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \tag{B.32}$$

$$TNC_{ijmr} = TNCF_{ijmr} + TNCE_{ijmr} + TNCI_{ijmr} \qquad \forall\,(i,j) \in E,\ m \in \mathcal{M},\ p \in P,\ r \tag{B.33}$$

RTI purchase.

$$\sum_{(i,j)\in E}\sum_{m\in\mathcal{M}}\sum_{p\in P} TNC_{ijmr} \leq s_r^{\text{RTI}} + P_r, \qquad P_r \geq 0 \qquad \forall\, r \in R \tag{B.34}$$

Daily transport costs.

$$TCS_{ijm} = \left(\sum_{r\in R} VS_{ijmr}\right) c_{ijm} + \left(\sum_{r\in R} X_{ijmr}\right) f_{ijm} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M} \tag{B.35}$$

$$TC_{ijm}\cdot q_{ijm} = TCS_{ijm} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M} \tag{B.36}$$

Logical Linking Constraints.

$$YF_{ijmr} \leq \sum_{p\in P} XF_{ijmpr} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M},\ r\in R \tag{B.37}$$

$$\sum_{p\in P} XF_{ijmpr} \leq |P|\cdot YF_{ijmr} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M},\ r\in R \tag{B.38}$$

$$YI_{ijmr} \leq \sum_{p\in P} XI_{ijmpr} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M},\ r\in R \tag{B.39}$$

$$\sum_{p\in P} XI_{ijmpr} \leq |P|\cdot YI_{ijmr} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M},\ r\in R \tag{B.40}$$

$$YM_{ijm} \leq \sum_{r\in R} X_{ijmr} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M} \tag{B.41}$$

$$\sum_{r\in R} X_{ijmr} \leq |R|\cdot YM_{ijm} \qquad \forall\,(i,j)\in E,\ m\in\mathcal{M} \tag{B.42}$$

$$XH_i \leq \sum_{j\in V}\sum_{m\in\mathcal{M}}\sum_{r\in R} X_{ijmr} \qquad \forall\, i\in H \tag{B.43}$$

$$\sum_{j\in V}\sum_{m\in\mathcal{M}}\sum_{r\in R} X_{ijmr} \leq |V||\mathcal{M}||R|\cdot XH_i \qquad \forall\, i\in H \tag{B.44}$$

$$YEOUT_{ir} \leq \sum_{j\in V}\sum_{m\in\mathcal{M}} XE_{ijmr} \qquad \forall\, i\in V,\ r\in R \tag{B.45}$$

$$\sum_{j\in V}\sum_{m\in\mathcal{M}} XE_{ijmr} \leq |V||\mathcal{M}|\cdot YEOUT_{ir} \qquad \forall\, i\in V,\ r\in R \tag{B.46}$$

$$YEIN_{ir} \leq \sum_{j\in V}\sum_{m\in\mathcal{M}} XE_{jimr} \qquad \forall\, i\in V,\ r\in R \tag{B.47}$$

$$\sum_{j\in V}\sum_{m\in\mathcal{M}} XE_{jimr} \leq |V||\mathcal{M}|\cdot YEIN_{ir} \qquad \forall\, i\in V,\ r\in R \tag{B.48}$$